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Diversity of spatial relationships of the claustrum and insula in branches of the mammalian radiation

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Using digital images of stained sections, we conducted a comparative survey of the visible gross morphology of the insular cortex and the claustrum in 26 mammalian species, representing most of the major mammalian radiations. We observed several features that are conserved in the mammals that we examined, including the absence of expansion of the insular cortex into out-branching lobes, the adherence of the claustrum to the fundus of the rhinal sulcus and to endopiriform cellular regions, and several features that varied, including the colocation of the insular and claustrocortex, the cortical origin of the operculations that cover the insula, the locations of large accumulations of claustral cells, and the presence of thin extensions of groups of claustral cells. We suggest that the final shape of the claustrum is highly influenced by the expansion of two medially adjacent subcortical structures, the striatum and amygdala. We further suggest that the neighboring locations of the claustrum and insula may be due to a dependence of each of these structures upon proximity to olfactory cortex, rather than to any connectional relationships between them.

Keywords: brain evolution; claustrum; insula; mammalian brain specializations; cortical folding

Introduction

In the past decade, much interest has been generated by data and hypotheses suggesting that two long-neglected regions of the brain, the cortex of the human insula and the neighboring cell group known as the claustrum, are important in such cerebral functions as the generation of the sense of self[1] or of consciousness itself.[2]

What is the insula?

The term *insula* was initially used by Reil[3] to describe the buried "fifth lobe" of the human brain. The term originally referred to a gross anatomical feature and not to any functional specialization or cytoarchitectural features of cortex. But as functions became ascribed to the human structure, the term insula was applied to regions in other species that were considered to be similar in supposed functions and connections. This is particularly problematic in species having no buried lobe, or no lobes at all, for example, the lissencephalic rodent species of rats and mice that became a popular experimental species model.

How the insula is identified in these animals varies, but identification is often reliant on the presence of the claustrum deep to the proposed insular cortex. The use of the claustrum to identify the insular cortex likely stems from the fact that the claustrum is found deep to almost the entire extent of the insular cortex in humans. However, this is not true of all mammals, as we shall demonstrate.

Connections of the insular cortex

As detailed in a recent review,[4] the mammalian insular region has strong reciprocal connections with anterior cingulate, frontal polar, dorsolateral prefrontal, temporal, and entorhinal cortical regions, as well as with amygdala and dorsal thalamus. It is the target of input from major sensory systems, including olfactory, taste, somatosensory,

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vestibular, auditory, and visceral sensory pathways.[1,5–10] It is also the source of efferent influences on visceral structures, including, notably, the laryngeal apparatus.[11] The presence of an abundance of connections with visceral organs, and with anterior cingulate and so many other cortical regions, has given rise to provocative ideas about this general region of the brain having a special relationship to awareness and control of internal bodily processes in relation to events in the exteroceptive milieu and to past experiences.[1,9,10]

Cytoarchitecture of the insular cortex

As a general statement, the cytoarchitectural subdivision of the insular cortex has not been easy to specify. A very detailed analysis of the region in rhesus monkeys[12] resulted in the popular picture of bands of progressively more elaborate cytoarchitectural lamination, starting with a simple single layer of cells in the antero-inferior corner, the "limen insulae," progressing posteriorly and superiorly through an "agranular" region of three or four distinct layers, then a "dysgranular" region with a spotty distribution of patches including more layers, and finally a "granular" region with the full complement of the isocortical pattern of six layers in the postero-superiormost region. This parcellation recognizes the absence of sharp cytoarchitectural boundaries and the transition from single-layered olfactory cortex bordering the monkey insula anteriorly to fully layered somatosensory and auditory regions on the posterior and superior insular borders. This scheme also poses a problem for classifying these insulas as allo-, meso-, or isocortex, since the different regions reflect characteristics of each of these traditional major subdivisions of mammalian cerebral cortex. Dogs, bats, hedgehogs, cetaceans, and artiodactyls do not have this pattern at all; their "insular" cortices are entirely or almost entirely agranular.[4]

Thus, cytoarchitectural distinctiveness is of little value in identifying the location or boundaries of a region considered the insula. It also renders attempts to describe "connections of the insula" quite confusing, since the differential architectures are likely related to different patterns of input to and output from the cells in a particular patch. At this time, we are left with only the folding pattern of the cortical lobes as a meaningful definition of the insula and insular cortex.

What is the claustrum?

The claustrum is present in almost all mammals, the exception being the monotreme platypus.[13,14] It consists in a homogeneous-appearing group of cells subjacent to cerebral cortex near the junction between isocortex and allocortex, as this junction passes near the underlying putamen in the vicinity just posterior to olfactory cortex.

Claustral anatomy

Historical descriptions of the claustrum are unclear about its boundaries,[15] notably between claustrum and overlying cerebral cortex. Recent proteomic studies in the rat and the green monkey (*Chlorocebus sabaeus*) have revealed that the claustrum is composed of a core of cells with distinct protein expression pattern different from that of cortical layer 6 neurons. This core is peppered with a sparse population of neurons displaying cortical protein expression.[16]

The boundary between the claustrum and the endopiriform cell group, once called the dorsal and ventral claustrum, respectively, is also indistinct. A common feature of virtually all mammalian claustrums is the continuity, as seen in the Nissl and myelin stains that we use, with endopiriform cells. This continuity takes place around the fundus of rhinal sulcus in gyrencephalic mammalian brains. This led to earlier classifications of endopiriform cell groups as subdivisions of claustrum, but more recent studies of connections and cytochemistry show that they should be considered as distinct cell groups.[17,18] Despite these distinctions, the continuity, which we shall term the endopiriform-adjacent "root of the claustrum," persists in all the mammals that we observed, and constitutes about all there is to the claustrum in the monotreme echidna,[14] the only member of the monotreme subclass that has been shown to exhibit a claustrum.[13] For the current study, we use the fundus of the rhinal fissure as an arbitrary boundary between endopiriform nuclei and claustral cell groups, following the example of Wojcick *et al.*[19]

Claustral connections

The claustrum has widespread and reciprocal connections with much of neocortex, and this characteristic, among others, caused Crick and Koch[2] to suggest the claustrum as the seat of conscious perception. The connectivity of the claustrum has

been traced extensively in the cat, macaque, rat, and rabbit and includes connections with primary somatosensory cortex; somatosensory association areas; primary visual cortex and peristriate cortex; auditory association cortex; primary motor, premotor, and supplementary motor cortex; cingulate cortex; the parahippocampal gyrus; temporal pole; and areas of prefrontal and orbital cortex.[2,20,21] Claustral connections are organized topographically, and specific areas of the claustrum broadly correspond to specific sensory cortices. Some claustral connections with subcortical structures have been reported, including connections with the intralaminar thalamic central lateral nucleus, the lateral geniculate complex, the lateral hypothalamic area, the basolateral amygdaloid nuclei, and the nucleus basalis.[21] However, the finding by Mathur *et al.*[16] of cells of the insular cortex surrounding the claustrum may offer an explanation for some of these projections. Whether these subcortical projections arise from the claustrum proper or from displaced cortical cells has yet to be resolved.

The claustrum as related to the insular cortex

A major difference among many species is the degree of distinction of claustrum from the overlying cortex.[22] Many animals show a conspicuous extreme capsule separating claustral cells from cortical cells; in others, the claustrum lies directly beneath or perhaps within the cortical layers.

The origin of the claustrum has been the focus of some debate; many early neuroanatomists believed that the claustrum was a derivative of the insular cortex and this may be the original source of the use of the claustrum to identify the insula. This argument would support the use of claustrum as an identifying feature for the insular cortex in animals without large brains and the expansion of operculum that accompanies them. Other researchers argued that the claustrum has a subpallial origin, because of differences in cytoarchitecture compared to cortex and examples of insulaless claustrum in pathological cases. More recent studies suggest an intermediate origin.[20,23]

Previous surveys of claustral morphology across several species include those for the shrew, rat, mouse, guinea pig, rabbit, macaque, vervet monkey, and human,[24] and for the walrus, hippopotamus, bongo, manatee, and beluga and Minke whales[4]—a total of 15 species and 9 mammalian orders. In this study, we have examined the claustrum and insula of 26 species of the mammalian radiation, representing 12 orders, and all of the major superordinal branchings except the monotremes.

We have found that the claustrum can be present in areas of the cortex besides the insula, and that the insulation of cortex, the anchoring of specific areas while other areas develop around it, is not always concurrent with the presence of the claustrum. We have identified several common patterns of claustrum formation, and we identify some common features that may lead to the claustrum taking these different patterns. We highlight several unexpected and interesting claustral morphologies, and we explore implications for theories of cortical expansion.

Method

As an example of investigations that can be carried out using these valuable and available resources, for this project, we relied entirely on digital images from three brain collections, assembled together at the National Museum of Health and Medicine thanks to the efforts of Wally Welker. Most of these images are available online, also thanks to Welker's efforts and personal activity.

In our study, we followed Wally Welker's principle[25–27] that scientific investigations should begin with a careful and comprehensive observation of what exists in the natural state that has been produced by genetic, ecological, and evolutionary processes. Accordingly, before resorting to experimental and analytical approaches, we undertook here natural observations of the situation of these brain structures in the variety of diversification that they have achieved in the progress of mammalian adaptation through evolutionary time and events.

For such observations, we selected brains of species that represent the major branches of the mammalian radiation, for which we had available stained section series. In our depictions of these section series, certain structures have been labeled: olfactory cortex, insular cortex, claustrum, and putamen. Insulas were identified as expansions of rhinal sulcus that become operculated in their posterior extent, usually accompanied by the presence of claustrum deep to the overlying cerebral cortex. The species that were selected are listed by abbreviated common names in Figure 1, along with their phylogenetic relationships based on the tree

presented by Springer *et al.*[28] The currently used complete common and scientific names of each species are shown along with the pictures of stained sections in Figures 2–8. The names used are those used in *Mammal Species of the World,*[29] as cited in the website established by Wally Welker: Mammalian Brain Collections at http://brainmuseum.org.

The sections that we used are from the Welker–Wisconsin, Yakovlev–Haleem, and Johnson–Michigan State brain collections. All sections are from brains that have been dehydrated and embedded in celloidin, sectioned, and then stained with thionin or cresyl violet, Nissl method, to show cell bodies, or with iron hematoxylin, one or another of variations of the Weigert method, to show myelinated fibers. To produce our figures, we selected the most anterior section that contained claustrum, the most posterior section that contained claustrum, and two evenly spaced sections between them. The analysis and comparison were conducted using the full sets of available sections, not just those illustrated here.

Results

Figures 2–8 present coronal sections through the claustral and insular regions from each of 26 species, at each of four sections approximately evenly spaced through the antero-posterior extent of these regions. In each figure, the anteriormost section is placed at the top of the page, the posteriormost section at the bottom. In the upper left corner of each page is a photograph of an intact brain from that species; in most cases, this is the same brain that was the source of the sections. White vertical lines on the photograph show the approximate level of each of the four sections.

At the lower left, the phylogenetic tree of Figure 1 is reproduced, with the name of the species whose

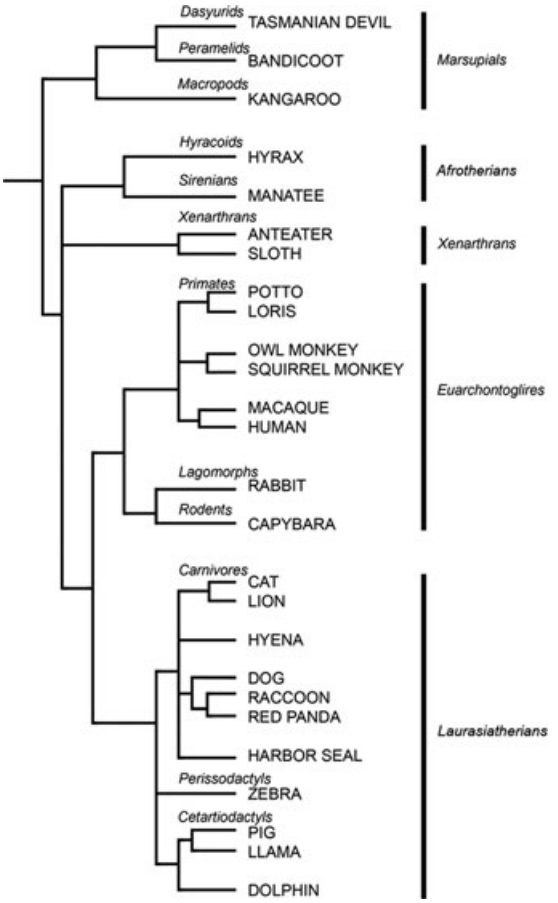


Figure 1. Phylogenetic relationships of species whose brains are depicted in Figures 2–8.[16]

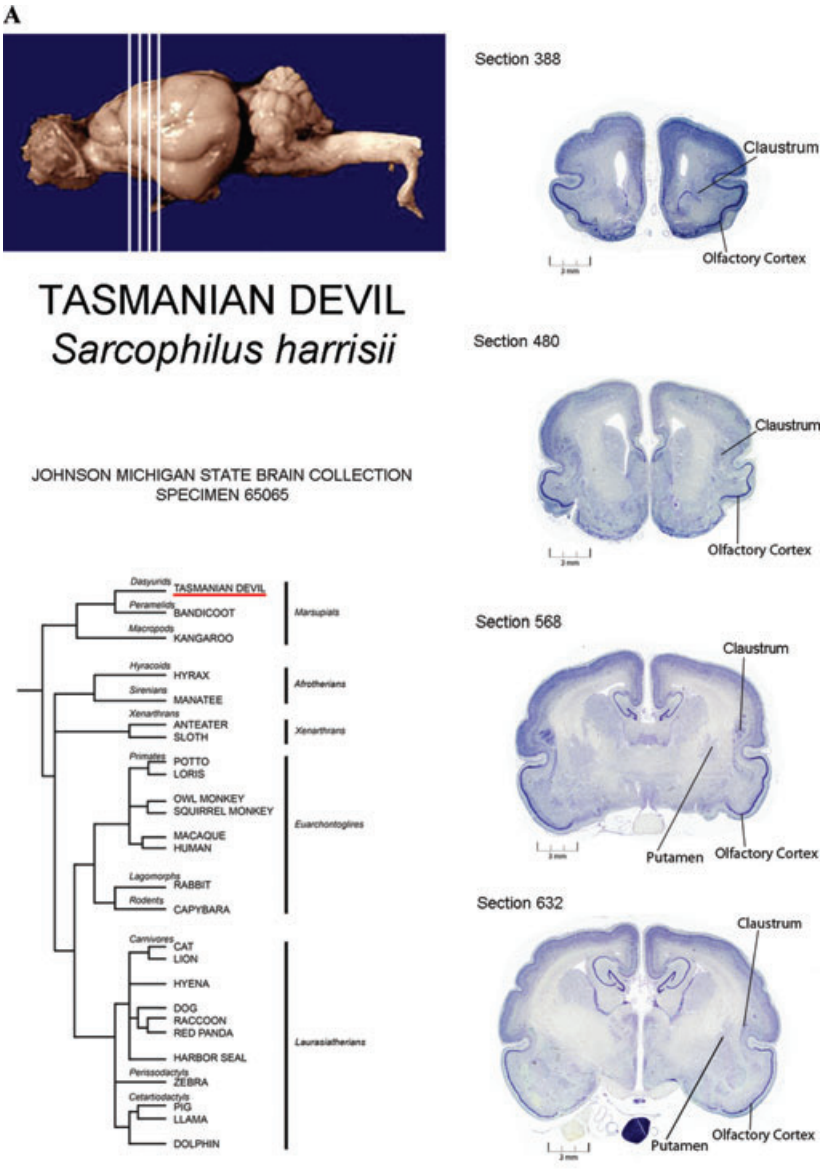


Figure 2. **Brains of marsupial specimens. (A) Note the large clump of claustral cells in sections 480 and 568 and the "root" of the claustrum reaching into the endopiriform region in sections 568 and 632.**

brain is displayed on that page underlined in red, so as to identify where in the tree the species is located and its relationship to the other species used in this study.

Marsupials

We examined three mammals of the marsupial lineage, a carnivorous Tasmanian devil, a small omnivorous bandicoot, and a large herbivorous kangaroo, each representing a large ordinal group. Each of these brains is characterized by a lack of a corpus callosum and an expanded anterior commissure and hippocampal commissure, as are all marsupial brains (Fig. 2).

Insula. None of the marsupials we have examined have an operculated insula. Both the kangaroo and the Tasmanian devil have rhinal sulci that are flanked by claustral and endopiriform cells.

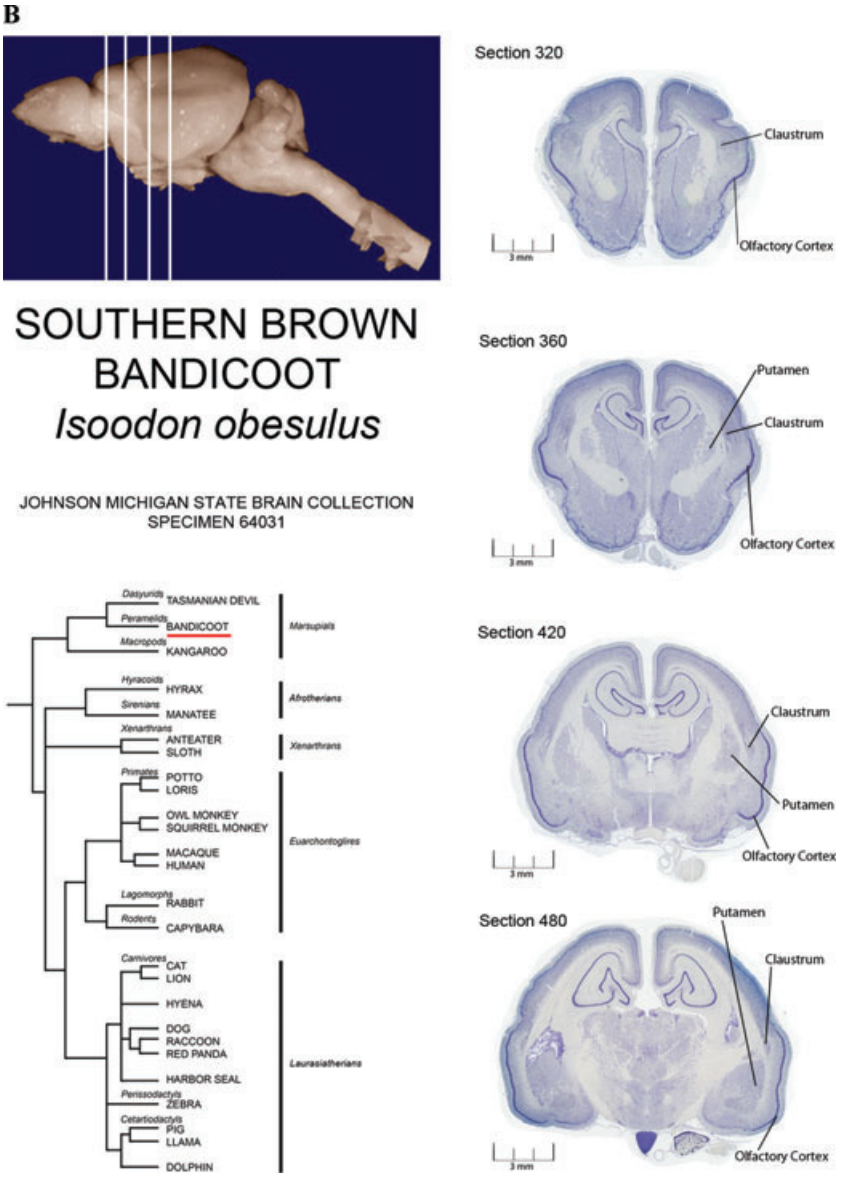


Figure 2. *Continued*. (B) **A wide clump of claustral cells can be seen in sections 320 and 360, and a claustro-endopiriform root in sections 360 and 420.**

Claustrum. The brains of the marsupials in our set surprised us with their claustral shape, at odds with what our preconceptions about what a claustrum should look like: a laminar structure closely applied to a striatal rim and a cortical cover. The claustrums of marsupials are, instead, a simple anteriorly placed ball-like collection of cells in continuity with endopiriform cells, flattening into a more laminar shape posteriorly, still in contact with endopiriform elements. The placement of most of the claustrum anterior to the putamen may be a result of the large size of the anterior commissure in these marsupials, displacing the claustral cells to a relatively more anterior location.

Afrotherians

The recently recognized afrotherian radiation, separate from all of the other placental mammals, features extremes of variations in brains, bodies, and behaviors, as its members adapted to an unusually

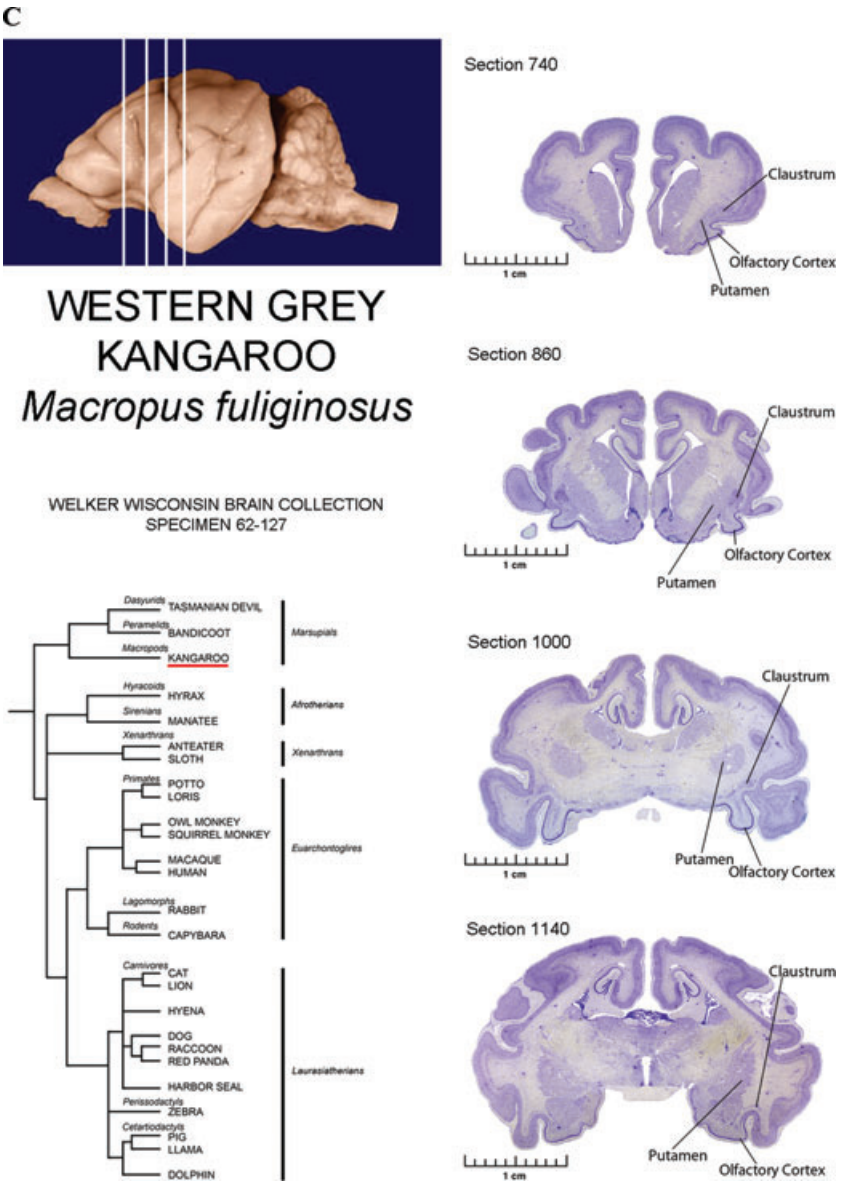


Figure 2. *Continued*. (C) **A broad group of claustral cells is found in sections 740 and 860, and a claustro-endopiriform root in section 1000.**

wide variety of niches on the isolated African continent. We examined two of its contrasting species, a moderately small-sized terrestrial, running and jumping animal—a hyrax; and a very large, fully aquatic, slow-moving one—a manatee. Their brains also exhibited maximal variations, showing unexpected and very unusual features in our regions of interest (Fig. 3).

Insula. The manatee brain is remarkably lissencephalic, strange for a brain of this size (Fig. 3B). There is one large fissure separating the brain into large anterior and posterior lobes. The lobes form operculations over a region of cortex that may be claustrocortex; this cannot be securely determined in the Nissl-stained sections that we had available. This region might be considered as

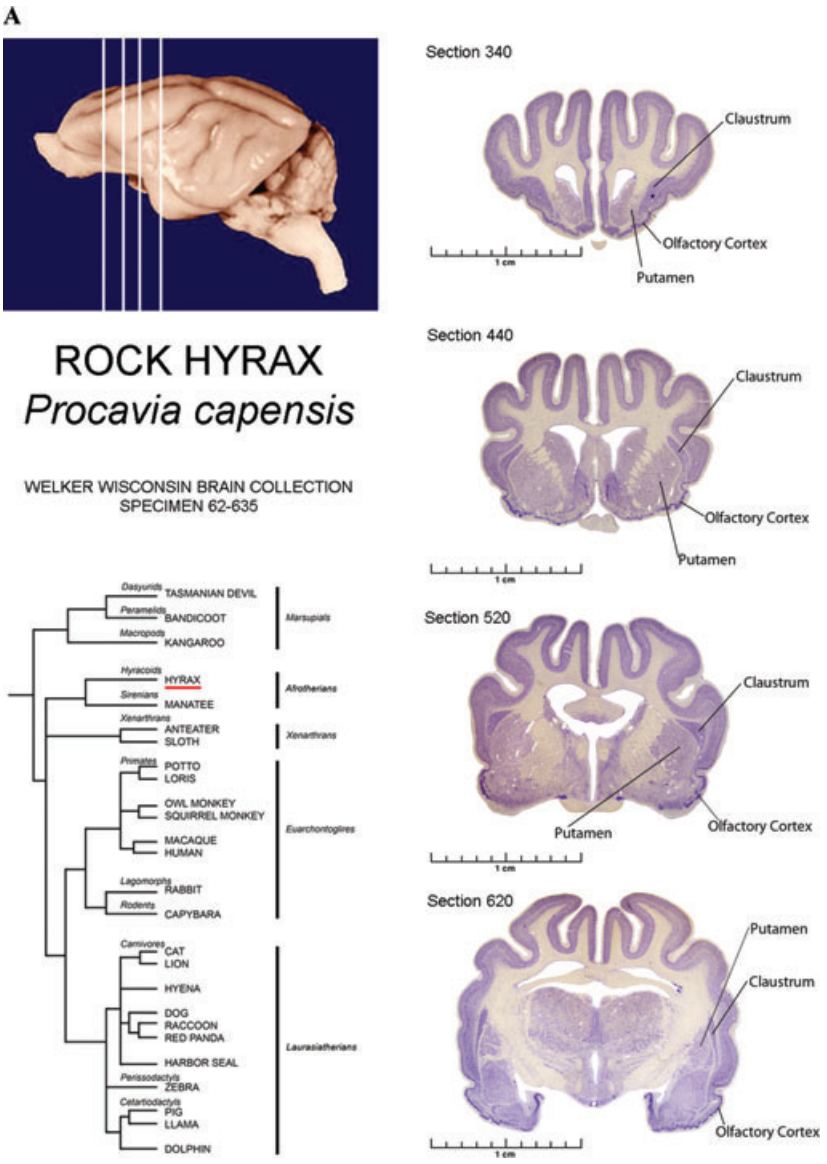


Figure 3. **Brains of afrotherian specimens. (A) An angled laminar claustrum in section 440 becomes triangular in shape in section 520. In these two sections, the claustrum underlies and projects into the white core of a gyrus with the beginnings of operculums overlying it on either side, a sort of "beginning stage" of insulation. In sections 520 and 620, a long thin claustral root extends around the fundus of the rhinal sulcus into the endopiriform region.**

insular using a claustrocortical-plus-operculated criterion.

The hyrax cerebrum is just as remarkably gyrencephalic for a brain of its small size (Fig. 3A). There is a small, shallow rhinal sulcus; superior to this is a sizable deep sulcus, the first of three parallel sulci that subdivide the superior aspects of the hemisphere. The cortex between the rhinal and the first deep sulcus is almost entirely claustrocortex, giving the gyrus a claim to be considered as a kind of proto-insula based on a sulcus-bounded claustrocortical criterion.

Claustrum. The hyrax offered a robust, broad, prominent, and well-defined claustrum, in contrast to a wispy, tenuous, almost vestigial claustrum

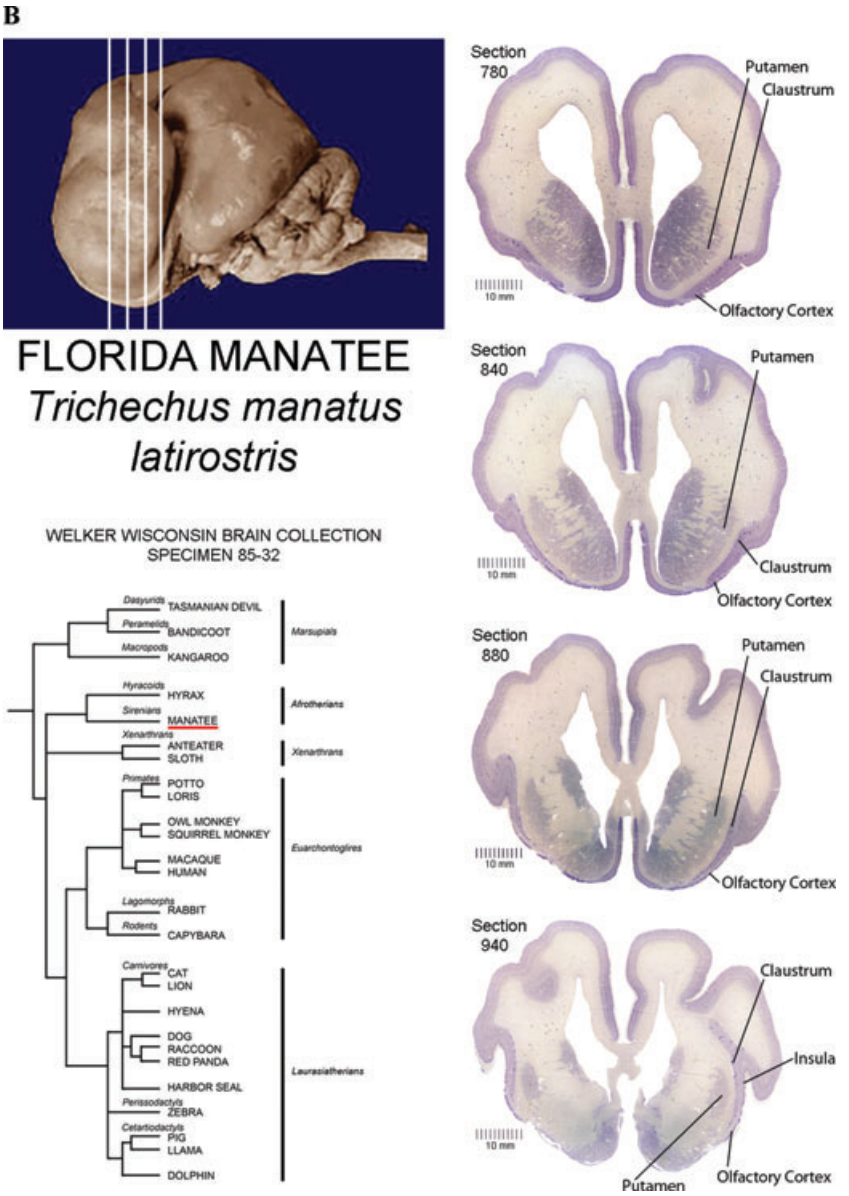


Figure 3. *Continued.* **(B) A thin, indistinct, and discontinuous line is seen at the inner margin of the cortex just superior to the olfactory cortex, in every section. This could be the claustrum. The upper margin of this putative claustrocortex is, in sections 880 and 940, the site of a folding forming the one large fissure of the cortex of the hemisphere; the claustrocortical side of the fold is buried under the other side, perhaps qualifying it as a sort of insulated claustrocortex.**

clinging to the internal margin of a ventrolateral region of cortex in the manatee brain.

The hyrax claustrum takes the form of a fairly thick sheet of cells, beginning anterior to the expansion of the unusually large putamen, and becoming a curved boomerang shape medially. In its anterior portion, the claustrum is not clearly distinct from cortex, but it becomes separate as the extreme capsule develops in the next several posterior sections. The claustrum here expands, becoming wider than the cortex of the funduses of the neighboring sulci. There is a very thin cell bridge reaching inferiorly

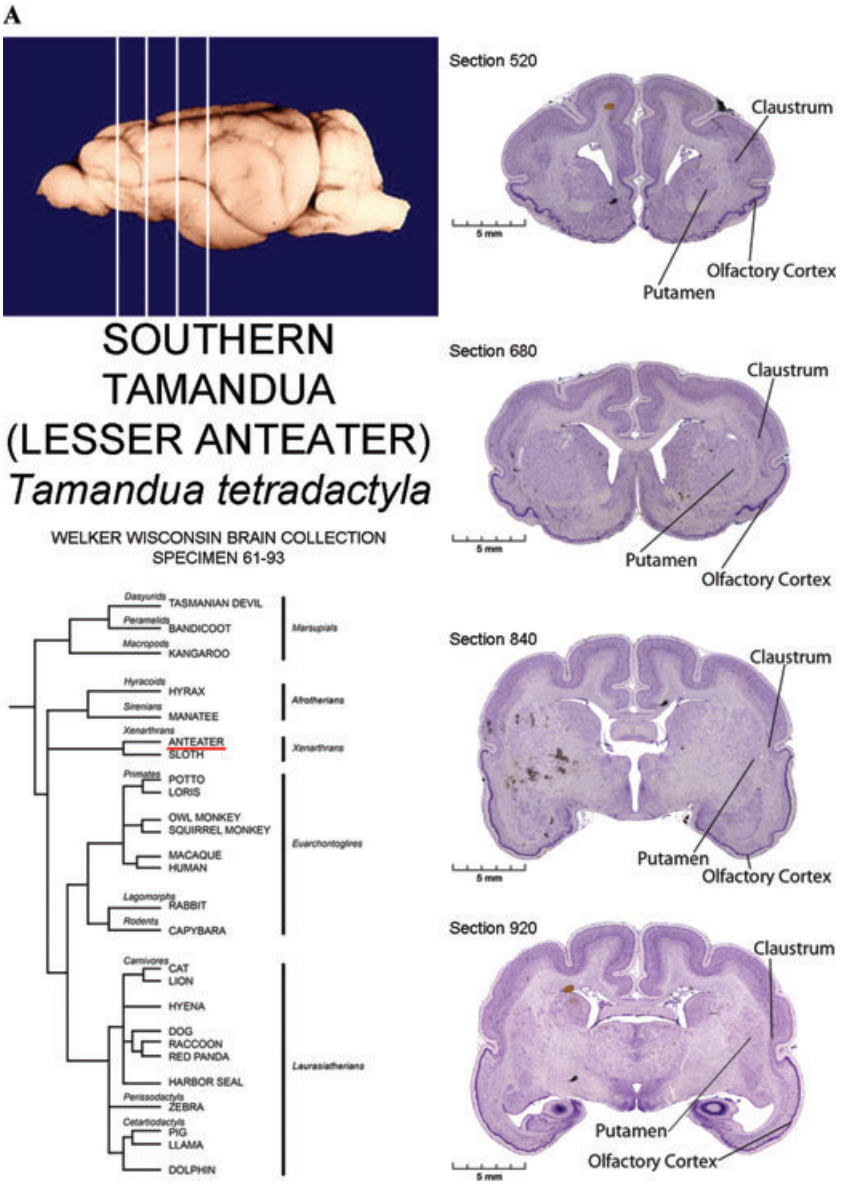


Figure 4. **Brains of xenarthran specimens. (A) The rounded claustral head lies in the anteriormost section 520, filling the center of a gyrus whose defining sulci show beginning operculations. This gives the gyrus the appearance of a proto-insula of the claustrocortex. The claustrum flattens caudally at its posterior end in stretching down past the rhinal sulcus as a claustral root reaching into the endopiriform region in section 920.**

to meet an endopiriform region, thus forming the "root of the claustrum" around the fundus of the shallow rhinal sulcus.

Like most of the manatee brain, the claustrum is very unusual. There is a thin, indistinct line of darkly staining islets of cells that lie in the deepest layer of cortex where a claustrum would be expected. As the extreme capsule is rudimentary or absent in these sections, it is impossible to say with any certainty that these cell masses represent the manatee claustrum. However, this deviation in claustral form warrants further investigation, and immunohistochemical techniques should be able to provide a definitive answer. Also missing is a clearly defined endopiriform nucleus. The darkly staining cellular islets are found deep to olfactory cortex, and may

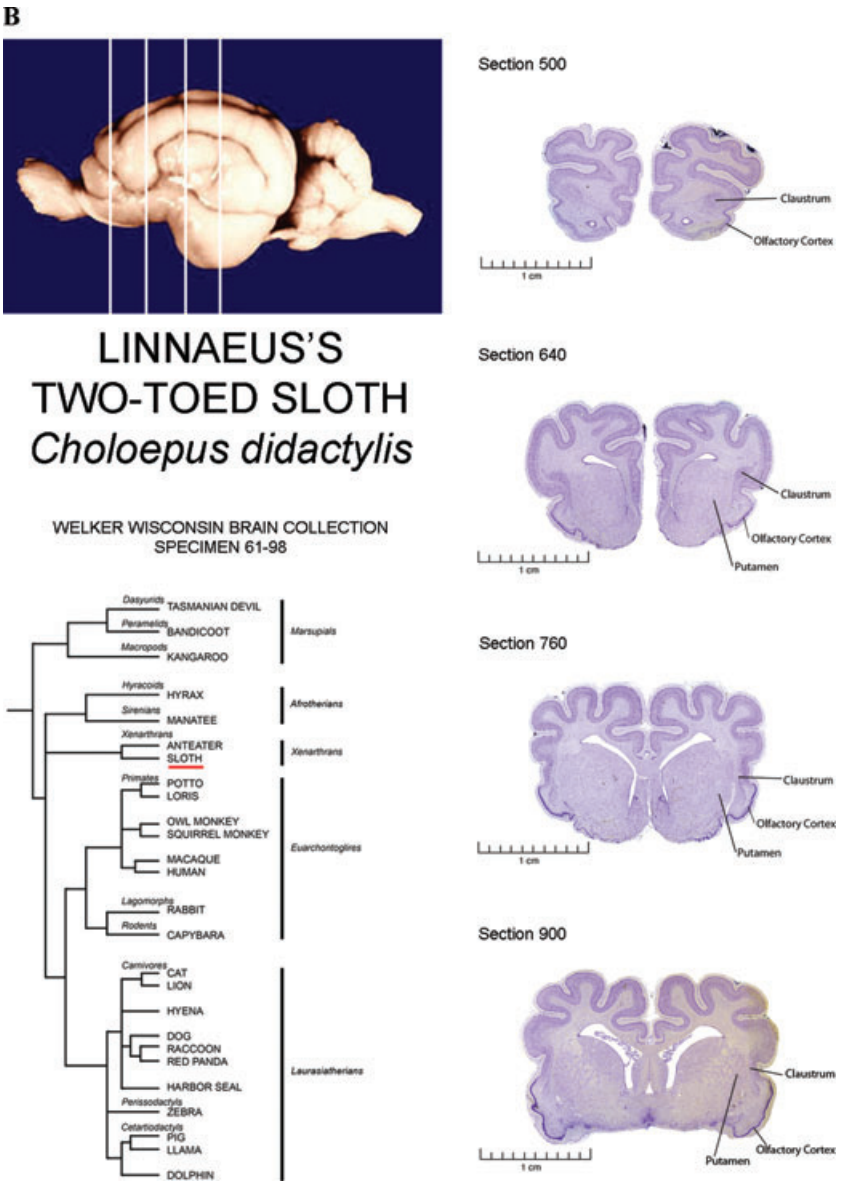


Figure 4. *Continued.* (B) Here again the rounded claustral head is anteriormost in section 500. In this case, the surrounding gyrus has a lengthy operculation by the next superior gyrus, leaving only the lower portion of this partly insulated gyrus edged with the claustrocortex. The claustral root into the endopiriform cells can be seen in all three of the remaining sections: 640, 760, and 900.

represent an endopiriform group; they also seem to be sporadically located over the full extent of the manatee's putative operculated insular claustrocortex.

Xenarthrans

Like the afrotherians and the antipodean marsupials, the surviving xenarthrans constitute a mammalian biodiversification occurring on an isolated continent—South America. Our specimens are an insectivorous terrestrial anteater and an herbivorous sloth that is specialized for arboreal navigation (Fig. 4).

Insula. Both of these animals have rhinal sulci flanked by claustrum. In the anteater, the rounded

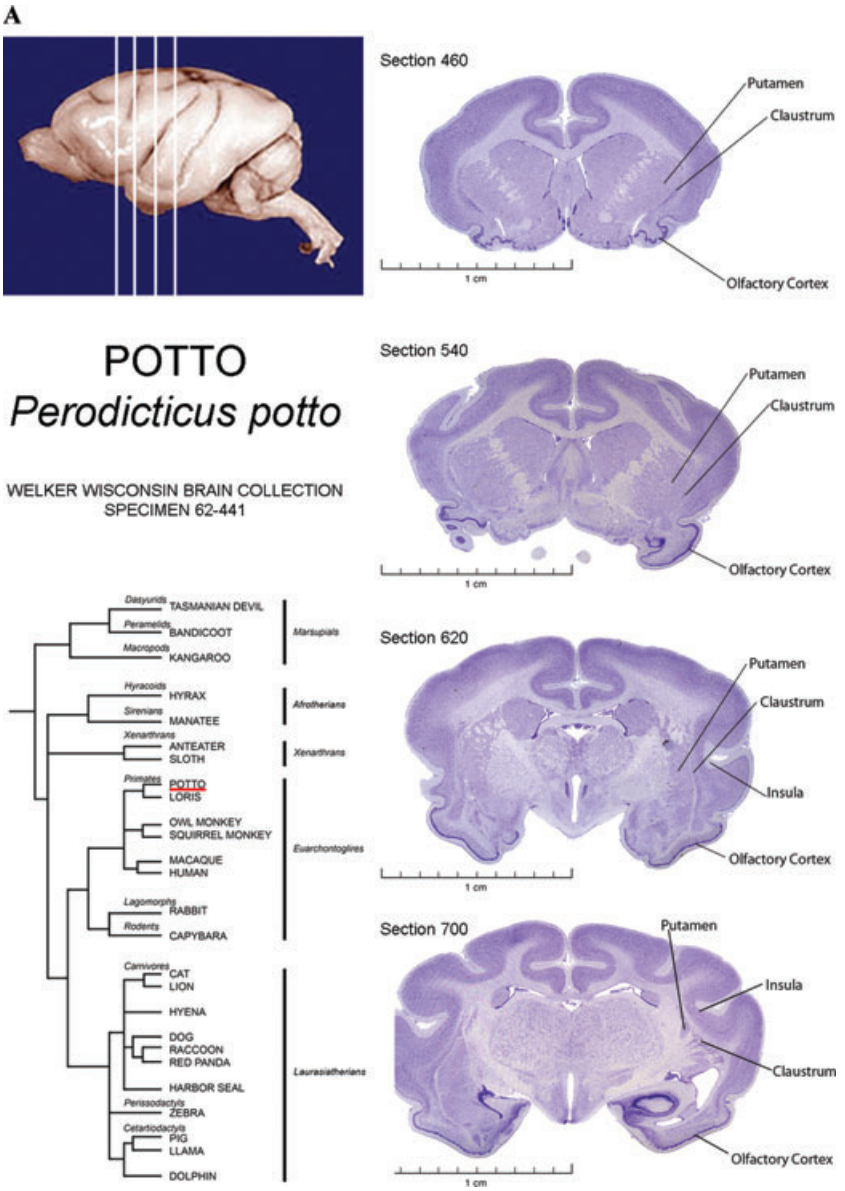


Figure 5. **Brains of primate specimens. (A) A thin laminar prosimian claustrum is seen in sections 460, 540, and 620. The claustral root to the endopiriform cells appears in section 620. Only scattered claustral remnants are visible in section 700. A gyral claustrocortex has buried portions, thus insulated, on its superior rim in sections 620 and 700.**

anterior head of the claustrum almost fills the core of a gyrus that might be considered a protoinsular claustrocortex, a configuration reminiscent of the midlevel of the hyrax claustrocortex.

Claustrum. These xenarthran claustrums look very similar to those found in the smaller of our species of marsupials. In each case, the claustrums form an anterior ovoid mass and flatten posteriorly between the expanding putamen and the superficial cortex. In both animals, the claustrum extends around the rhinal sulcus at every level, with a sort of tail extending posteriorly to meet endopiriform cells, thus forming a "claustral root."

Primates

The primate lineage is interesting not only because of our anthropocentric fascination, but also

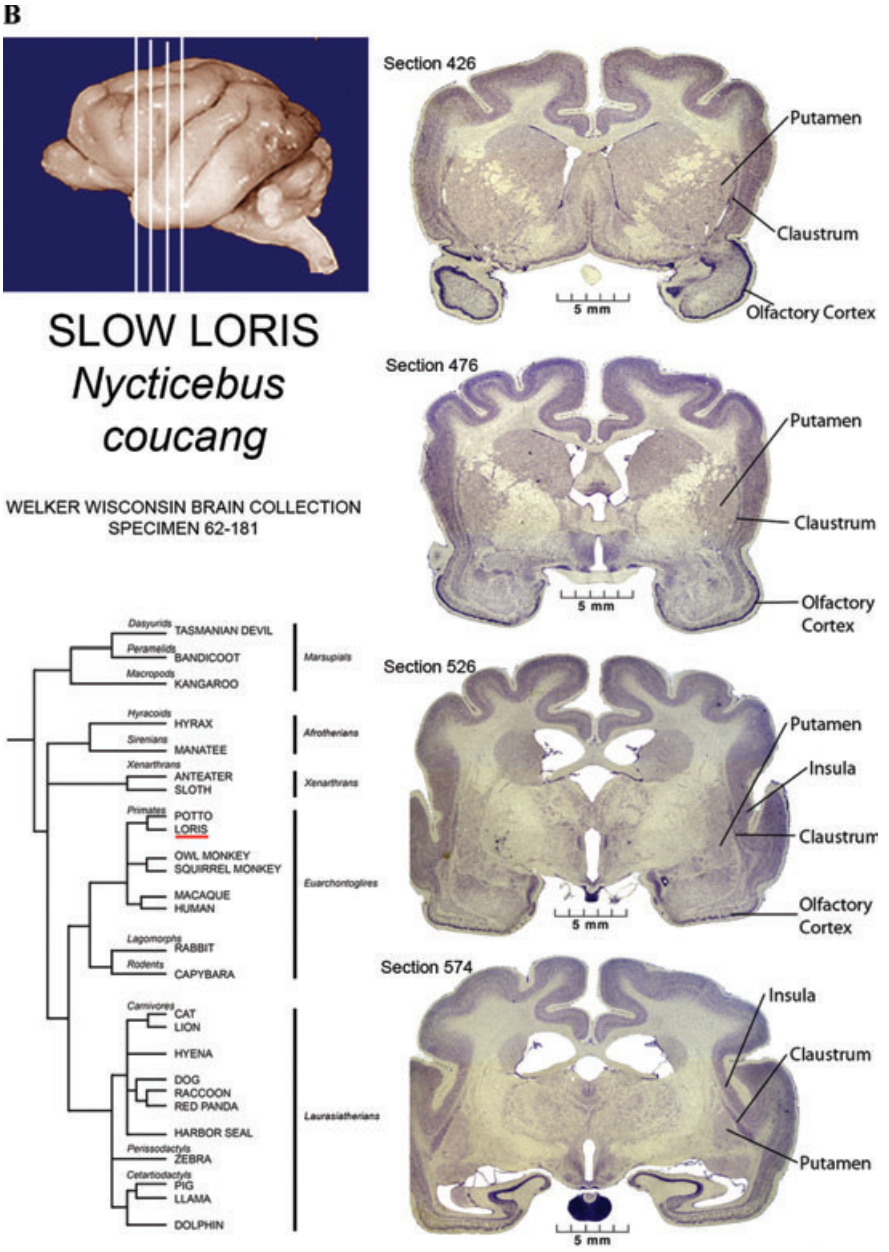


Figure 5. *Continued.* (B) Another thin laminar prosimian claustrum is evident in all sections, with the claustral root connecting to a substantial endopiriform cell group, next to the rhinal sulcus, in sections 476 and 526. In the posterior sections 526 and 574, a gyrus extends out from the lower part of the claustrocortex and folds over the upper part. That leaves the upper region of the claustrocortex deeply buried, that is to say, insulated by this gyral fold. The claustrum widens and extends into the white matter of the outfolding gyrus, a phenomenon that will be seen in a number of species to follow in this listing.

because it offers us a view of claustral and insular variation in a group of animals that have similar sensorimotor specializations but a wide variety of brain sizes. We examined two small prosimians (the potto and slow loris), two platyrrhines (the squirrel monkey and owl monkey), and two catarrhines (the pig-tailed macaque and the human) (Fig. 5).

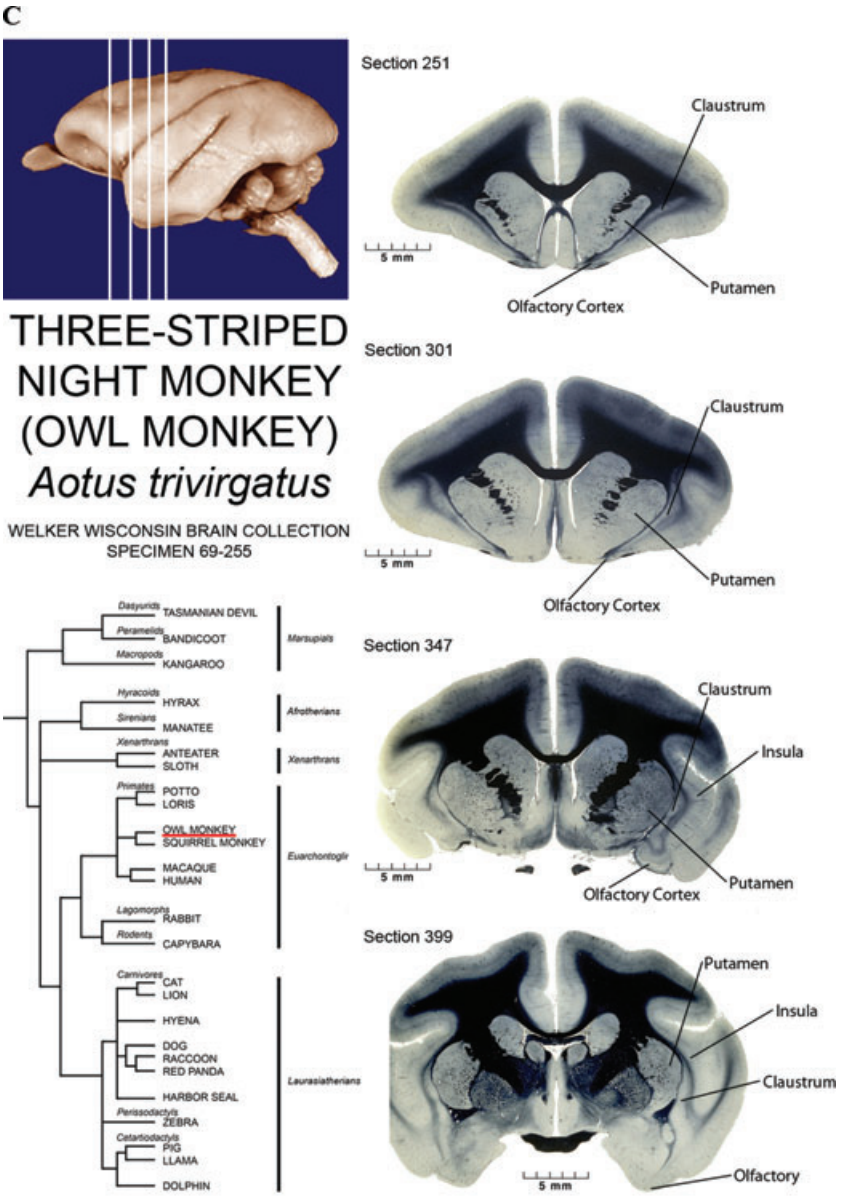


Figure 5. *Continued.* (C) In the brain of this platyrrhine New World monkey, the anterior end of the claustrocortex, in section 251, is uninsulated; complete insulation is seen in the posteriormost sections 347 and 399. The claustral root leading to endopiriform cell groups appears in section 347. These sections are stained for myelinated fibers by the Sanides–Heidenhain–Woelcke version of the Weigert iron hematoxylin method.[30]

Insula. All of our primates have insulas in some form. The primate insula is formed by a temporal operculum in all six species, and a frontal/parietal operculum in all of these except the slow loris. It is interesting that even the smaller monkeys have distinct insulas, while many much larger mammals do not. In the primate lineage, there seems to be a fairly robust correspondence between claustrocortex and insulated cortex. The insular cortex is almost completely claustrocortex, although the claustrum can wander from the insula. The human and macaque insulas are superficial to claustrum except in its most superior posterior portion.

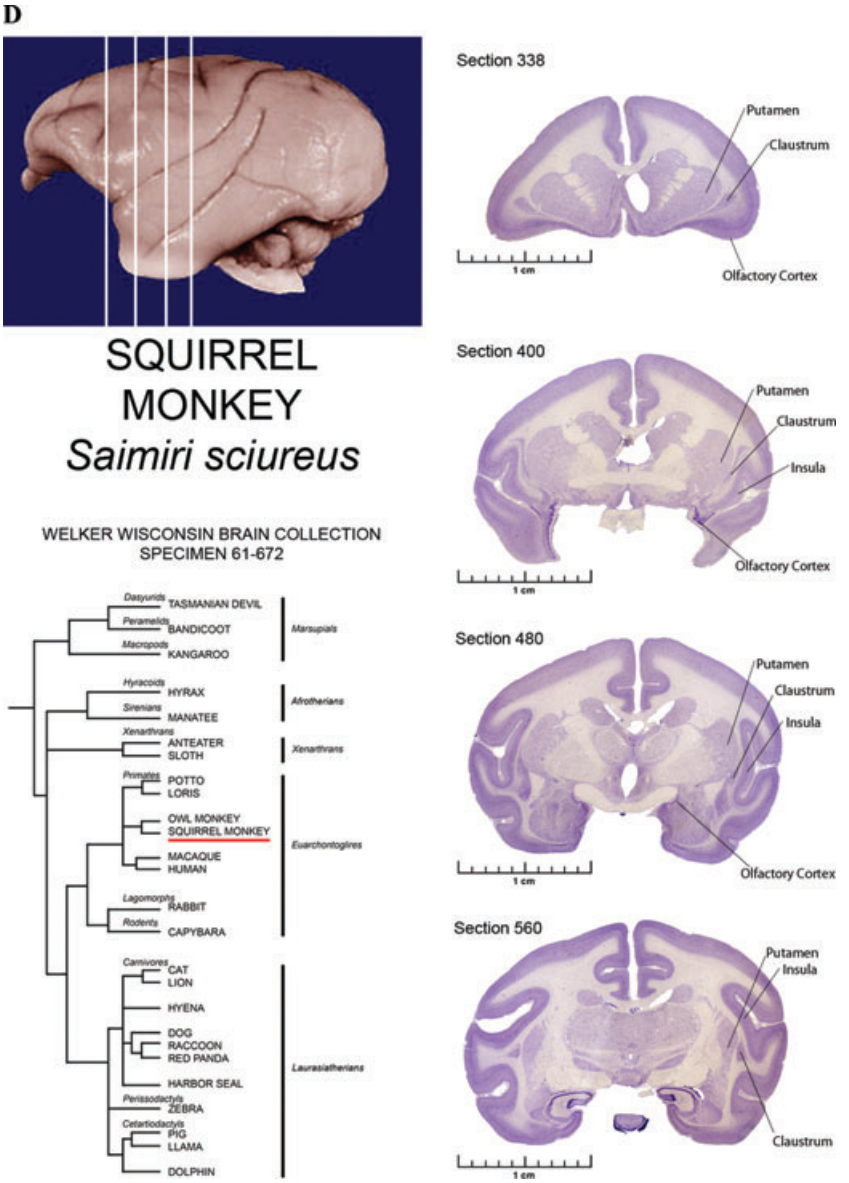


Figure 5. *Continued.* (D) Here, in another platyrrhine monkey brain, the anteriormost claustrocortex in section 338 is not insulated. In each of the remaining sections, complete insulation of the claustrocortex is evident. In section 400, a very short, but substantial, claustral root reaches endopiriform cells. Section 480 shows a broad inferior enlargement of the otherwise laminar claustrum, and in sections 480 and 560, the lateral tip of this enlargement reaches into the core of the inferior or temporal insulating gyrus.

Where the claustrum does stray from the insular cortex is where it reaches below the insula to become continuous with the endopiriform formation. In the owl monkey, we can see that the superior insula is claustrumless throughout its extent. In the squirrel monkey, we can see that the posterior superior insula is claustrumless, and the anterior claustrum is invading the frontal operculum. The potto's rather rudimentary insula is completely claustrocortex.

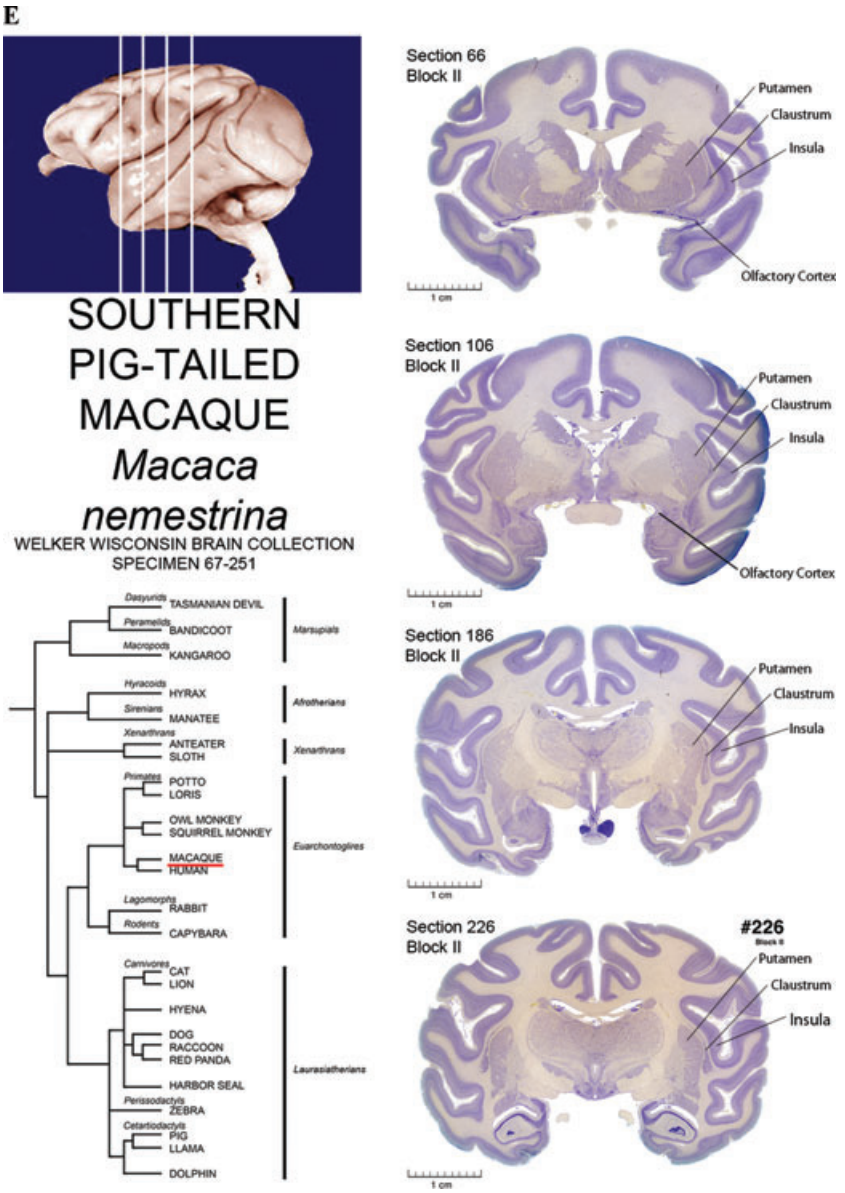


Figure 5. *Continued.* (E) This Old World catarrhine monkey brain shows the classical insulated claustrocortical arrangement. At the anterior end, in section 66, the claustrum continues on past the end of the insulated region of the cortex into the cortex of the superior operculating gyrus. At its postero-inferior end, the thin claustral lamina widens considerably in sections 186 and 226. The very thin claustrum-to-endopiriform root is seen in anterior sections 66 and 106.

Claustrum. In each of the six primates we examined, the primate claustrum takes the shape of a thin, vertically oriented flat sheet. Unlike the marsupials and carnivores, the external capsule of the primates is small compared to the size of the putamen, which is very large in the primate lineage and crowds the claustrum. This leads to the characteristic primate claustral form, a thin sheet of cells wedged between the putamen and the insula.

Also unlike the marsupials and carnivores, the claustrum does not appear as a distinct cell mass anterior to the level of the striatum. The putamen is large at the level where the claustrum begins,

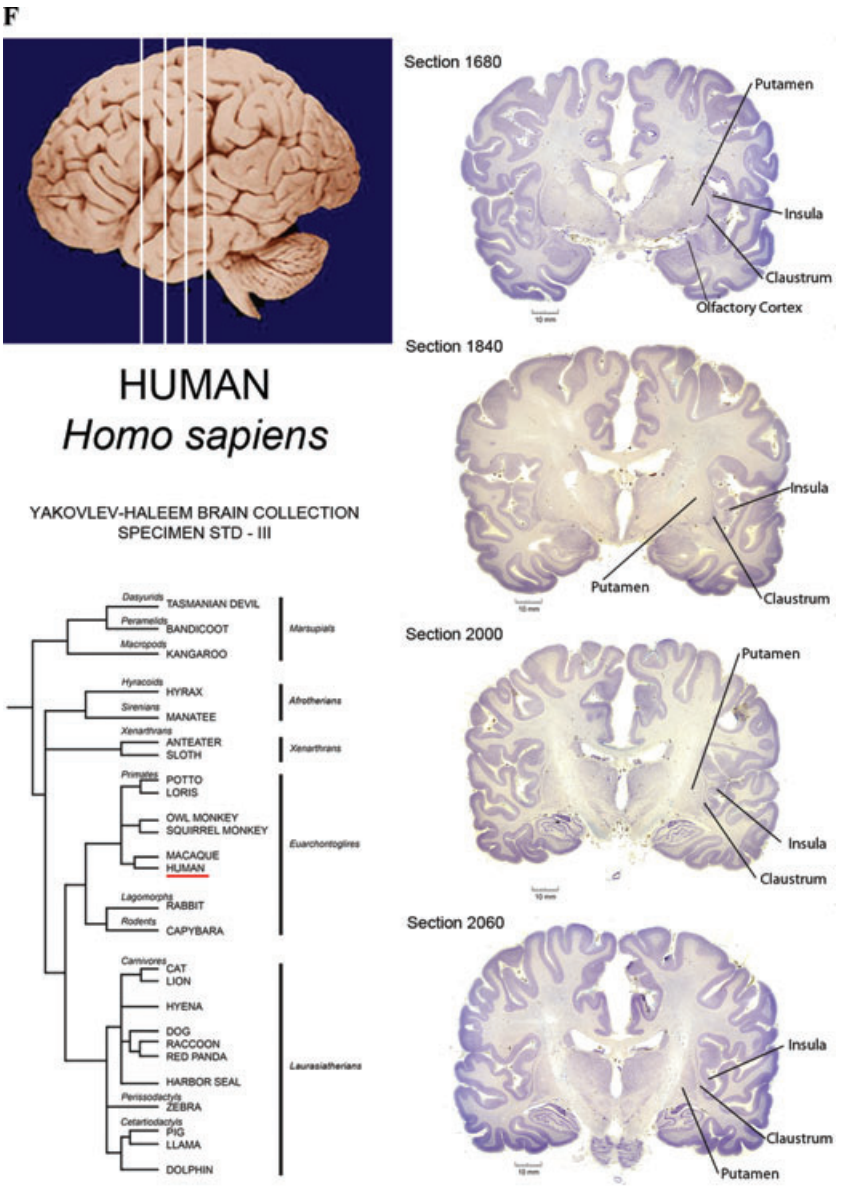


Figure 5. *Continued.* (F) **The human claustrocortex, as has long been recognized, is approximately coterminous with the insular cortex. A trace of a claustral root leading to a para-amygdalar endopiriform region can be seen in section 1680. In sections 1840 and 2000, the inferior portions of the claustral lamina widens noticeably, with extensions into cores of the most inferior insular gyri—these extensions in all species can be viewed as spaces outlined by U-fibers connecting neighboring insular gyri.**

perhaps impeding the development of a rounded anterior claustral mass, as seen in the marsupials, hyrax, rabbit, sloth, and anteater. In the large-brained primates, the human and the macaque, the anterior claustrocortex is operculated, while in the smaller-brained primates the anterior claustrocortex is found at the surface of the brain and is not insulated. This is likely to be the result of a relatively larger increase in the size of the insula compared to the size of the claustrum as total brain size increases in the primate lineage.

All of the primates we examined have a large, scimitar-shaped endopiriform cell group at the level of the anterior amygdala. Cell bridges connect this

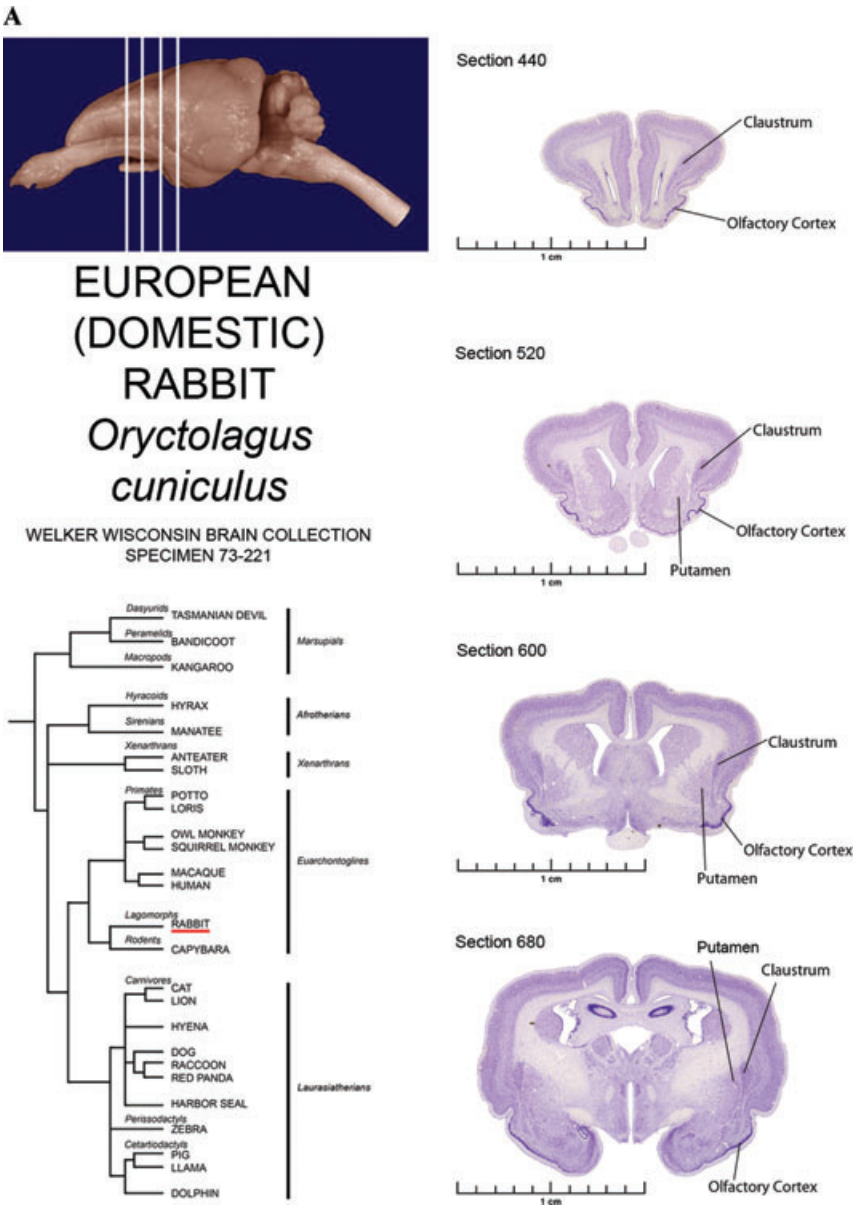


Figure 6. Rabbit and rodent brains. (A) In the rabbit brain, a distinct claustrum that occasionally shows boundaries consisting of sporadically visible extreme and external capsules. In each of our sections, the claustrum has a rounded club shape with its widest extent uppermost. At the inferior end of the claustrum, a thin trail of darkly staining cells forms a root of the claustrum, leading to endopiriform cellular regions. This is also seen in each of the anteroposterior section levels shown here.

structure to the claustrum. The primate endopiriform region features a prominent widening that could be due to its location inferior to the level of the putamen. In this area, the structure is unconstrained by the lateral border of the putamen and has room to expand. A similar effect can be seen in the anterior claustrum of the marsupials, which is anterior to the striatum, and forms an egg or club-shaped mass rather than a thin sheet and in the triangular superior claustral expansion in the carnivores. The endopiriform region was relatively larger and much more distinct in all of the nonhuman

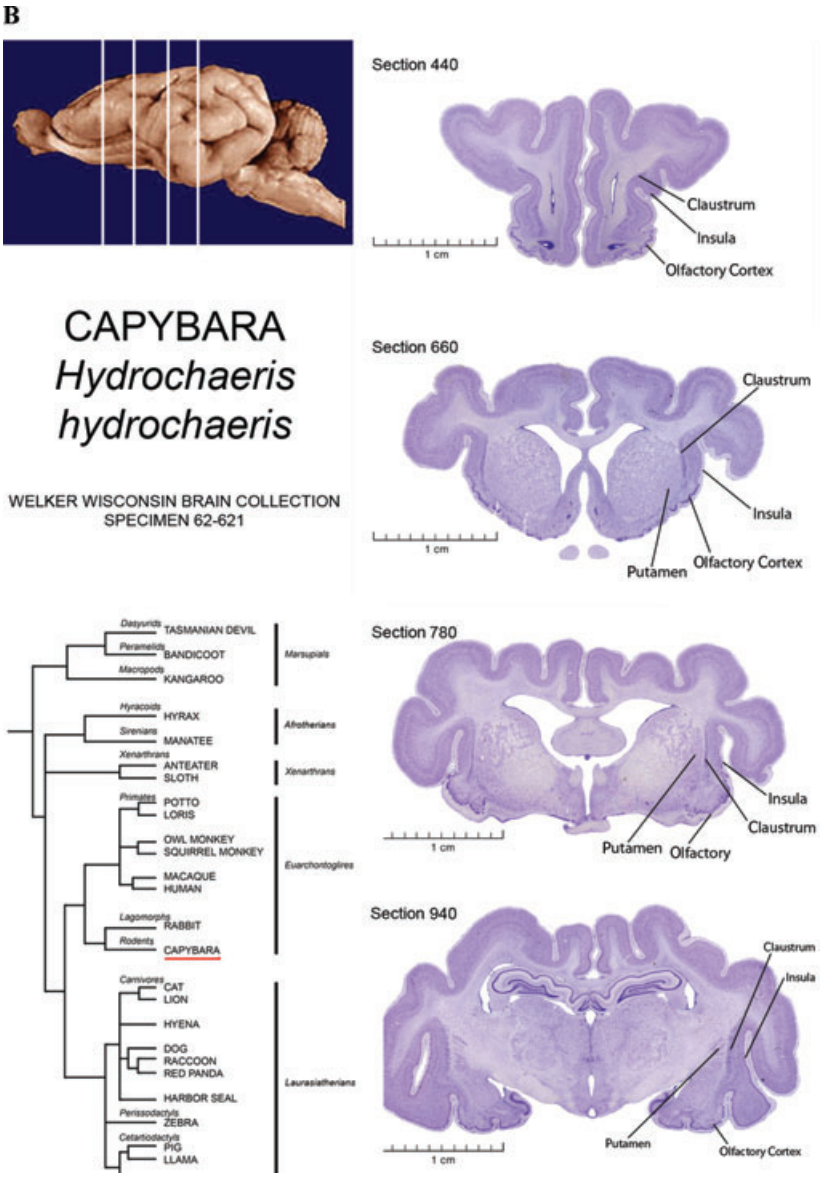


Figure 6. *Continued.* (B) The anteriormost section 440 through this large rodent brain features a pair of claustrocortical gyri, bounded by well-formed sulci, reminiscent of multiply gyrated insular formations seen in primates. At more posterior levels, only one claustrocortical gyrus is evident, but it is well operculated by a superior-lying gyrus folding over it, giving its status as insular formation. In sections 660 and 780, the rhinal sulcus disappears, unusual among mammals. A claustral root reaching around the rhinal sulcus and into the endopiriform region in the more posterior sections despite the disappearance of the rhinal sulcus is a constant feature in all of these sections. A fairly distinct external capsule and a less distinct but occasionally visible extreme capsule separate the laminar claustrum from the putamen and the cortex.

primates compared with the human (compare slow loris 476, owl monkey 399, squirrel monkey 480, and macaque 106 to human 1840). In the human, the endopiriform region is vestigial and this may be related to the decreased reliance of humans on the olfactory system.

Lagomorphs and rodents

These animals are representative of the clade Glires, which includes the orders Lagomorpha and Rodentia, believed at this time to be more closely related to one another than either is to any other mammalian orders (Fig. 6).

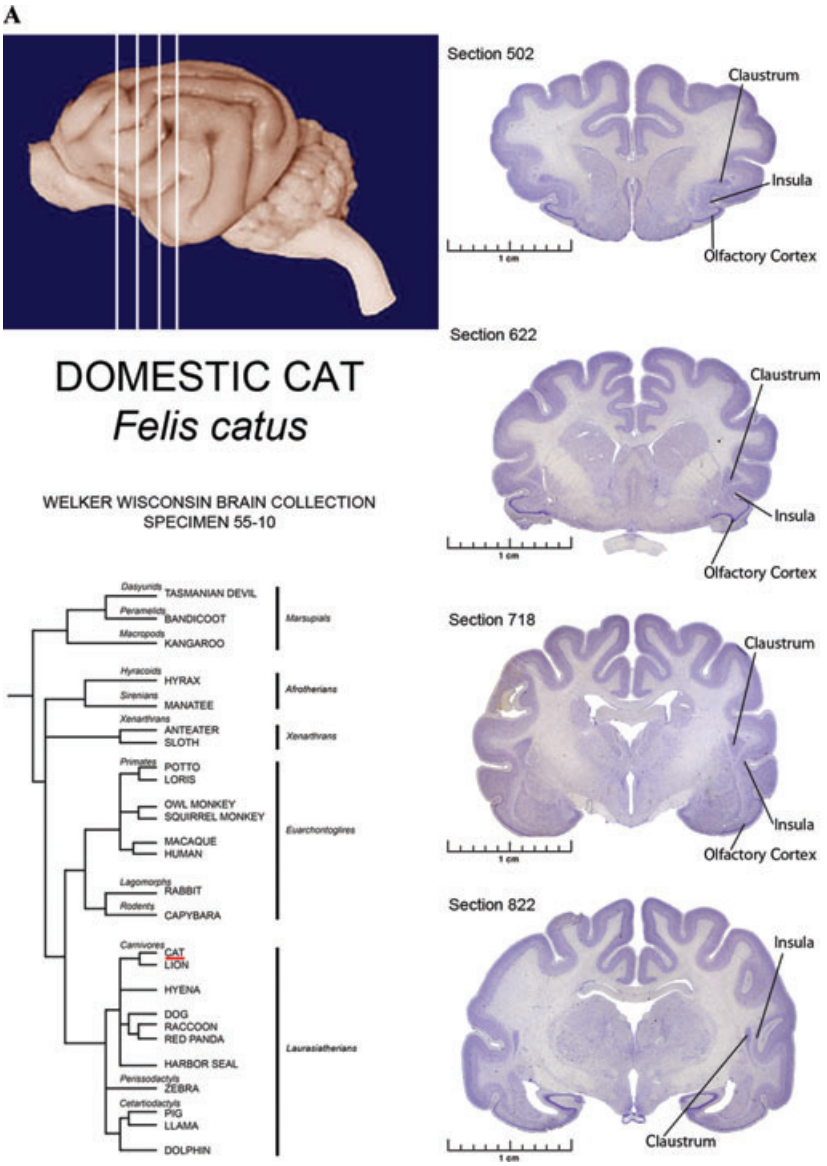


Figure 7. Carnivore brains. (A) This cat brain features, in the middle sections 622 and 718, a striking large triangular claustrum. The lateral apex reaches into the white core of a well-formed gyrus; the lower sulcal wall of the gyrus is closely approximate to the lateral side of the claustral triangle. The sulcus here in section 622 is the posterior end of the rhinal sulcus and the inferior apex of the claustral triangle leads into a claustral root reaching past the rhinal sulcus into the region of endopiriform cell groups. Medially, the claustrum is confined by the lateral side of the putamen. In the anteriormost section, 502, and the posteriormost section, 822, the claustrum flattens into a teardrop shape; the anterior shape is situated horizontally with the point of the drop facing medially; the posterior one sits vertically with the point reaching inferiorly. The cortex of the gyrus overlying the claustral triangle, particularly its lower bank, can lay claim to being a sort of proto-insulation.

Insula. Rabbits (Fig. 6A) show a rhinal sulcus and a couple of other very shallow grooves between gyral bulges. These shallow depressions do not form any type of operculations that would warrant calling them insulating.

Rodents, save for the capybaras and a few of their relatives, are almost completely lissencephalic; as such, they show no insulation. Our capybara specimens, large-brained and representing the most gyrencephalic of rodents, show gyral overfolding of

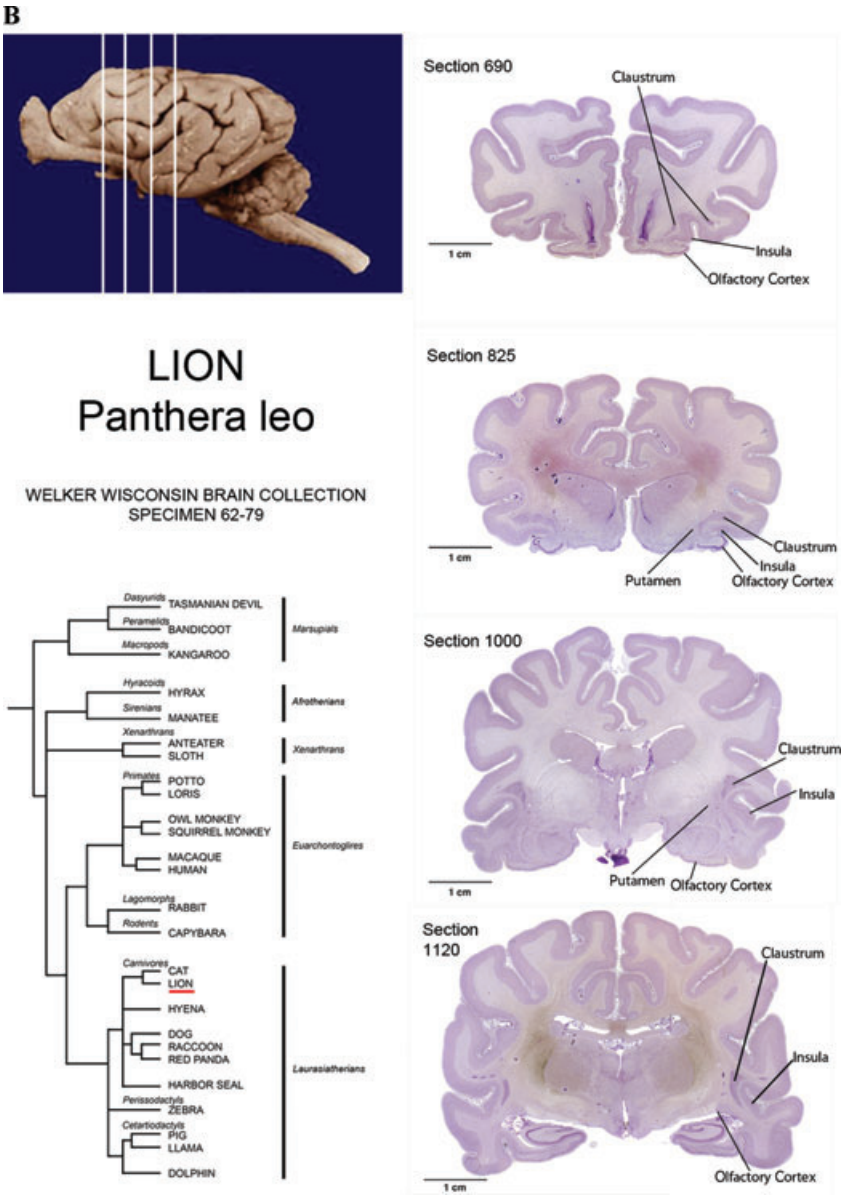


Figure 7. *Continued.* (B) The brain of the lion, a larger feliformian carnivore, in sections 1000 and 1120, shows a large triangular top to the claustrum similar to that in the cat, with the lateral apex of the triangle penetrating much farther into the overlying gyral core. Except for the absence of posterior flattening, the other aspects of claustral and insular morphology are very similar to those seen in the cat.

a claustrocortical region that could be called a proto-insula (Fig. 6B).

Claustrum. Our rabbit specimen (Fig. 6A) shows a substantial claustrum, club-shaped rather than a thin layer, appearing somewhat similar to the claustrum of our hyrax specimen. A clearly visible band of darkly staining cells extends downward into the endopiriform region; this root is particularly evident in the mid-level sections.

The claustrum of lissencephalic rodents is most similar to that seen in manatees among the animals so far considered: a thin layer of cells, beneath or within the deepest layer of cerebral cortex. In the

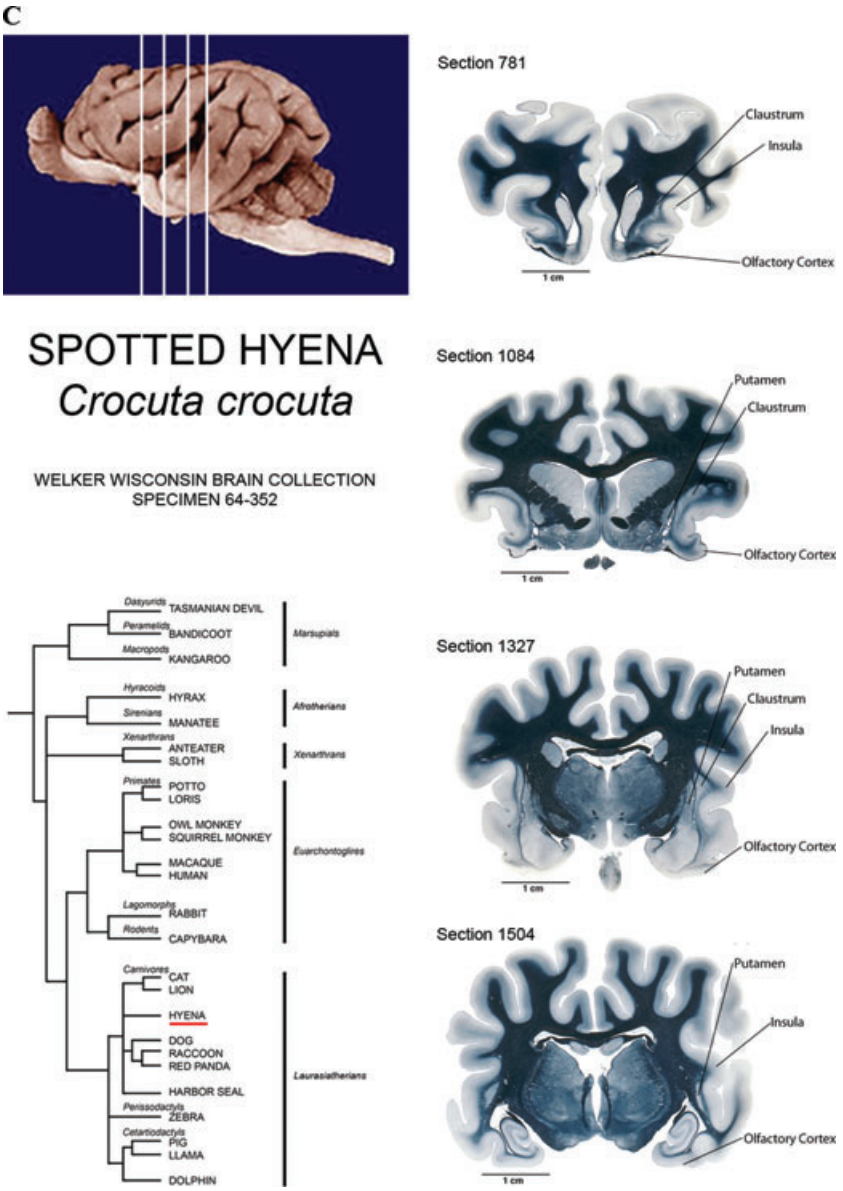


Figure 7. *Continued.* (C) **In the brain of our third feliformian carnivore, the "carnivoran triangle" of the claustrum, seen in section 1327, is less pronounced, and in section 1084 anterior to this, the wide top of the claustrum rounds up into a club shape. This rounding may be related to the lack of limitation by a bounding putamen this far forward. The very small putamen in the hyena is a pronounced example of a general phenomenon of reduction in putamen size among carnivores.[31] The other aspects of claustral and insular morphology are very similar to those seen in the cat and lion.**

substantially gyrencephalic capybaras (Fig. 6B), a readily visualized laminar claustrum lies between the partially insulated gyral surface and the putamen; thin external and extreme capsules can be seen on either side of it. Scattered cells at the inferior end of the claustrum may represent the claustral root in continuity with endopiriform cells.

Carnivores

Carnivores, in their extensive and varied radiation, show an interesting consistency in claustral morphology and a high degree of insulation in the larger-brained species. We studied the claustrum and insula of seven carnivores, including the cat, lion, hyena, dog, raccoon, red panda, and harbor seal.

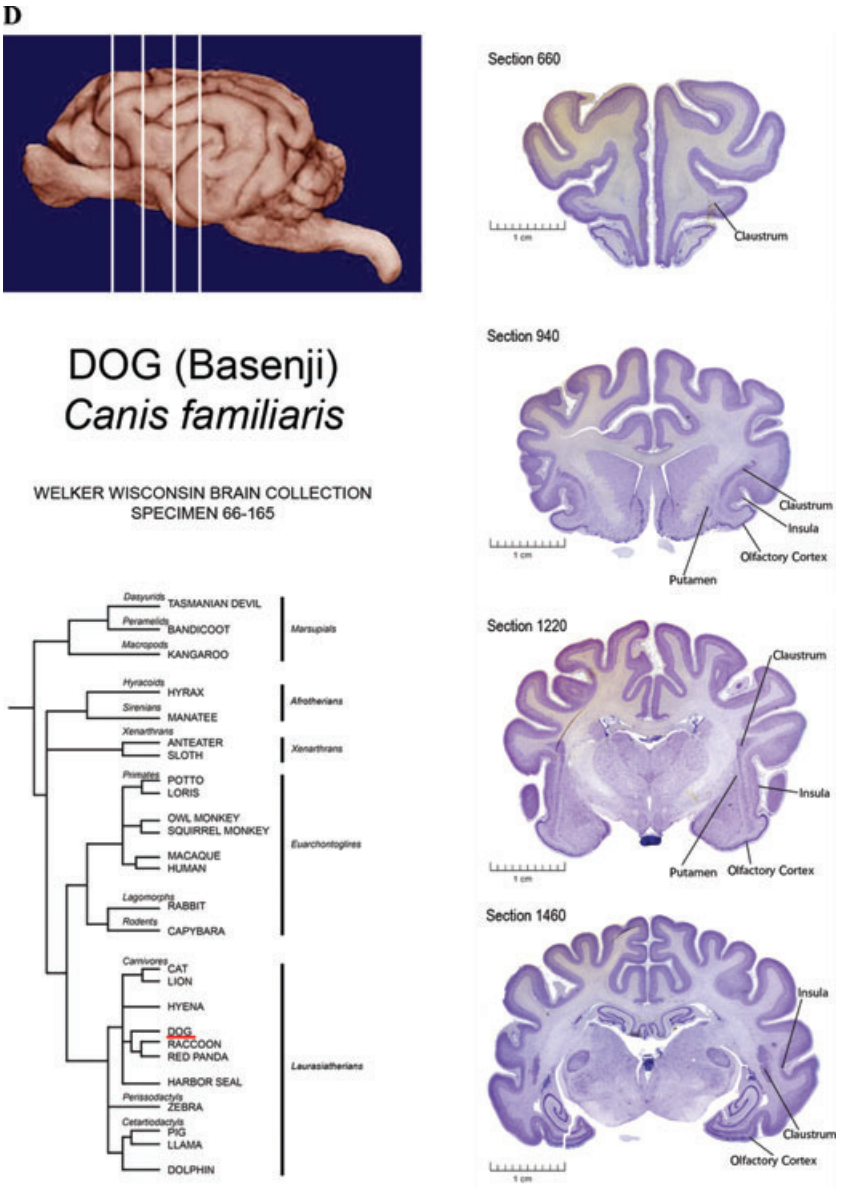


Figure 7. *Continued.* (D) **In the caniformian carnivore brain of this dog, the triangular swelling of the superior portion of the claustrum is seen atop a long claustral column in section 1220. The cortex alongside the column is buried under, or insulated by, an operculating gyral lateral outcropping. The inferior end of the column arrives at the rhinal sulcus and continues on as the claustral root into the endopiriform cells below.**

Insula. In most of the carnivores we examined, the insula begins anteriorly at a small rhinal sulcus, located on the underside of the brain. This is present in all the carnivore brains that we examined except that of the hyena. Posteriorly, as the amygdalar complex expands medial to the insula on the inferior surface of the carnivore brain, the insula is pushed to the lateral face of the brain. Here, the insula expands, and insular gyri appear in the larger brains. There is little continuity of form, however. Both a superior and an inferior operculum, for example, cover the small insula of our lion almost equally, while the harbor seal insula is formed by a great expansion of the inferior operculum, encircling the

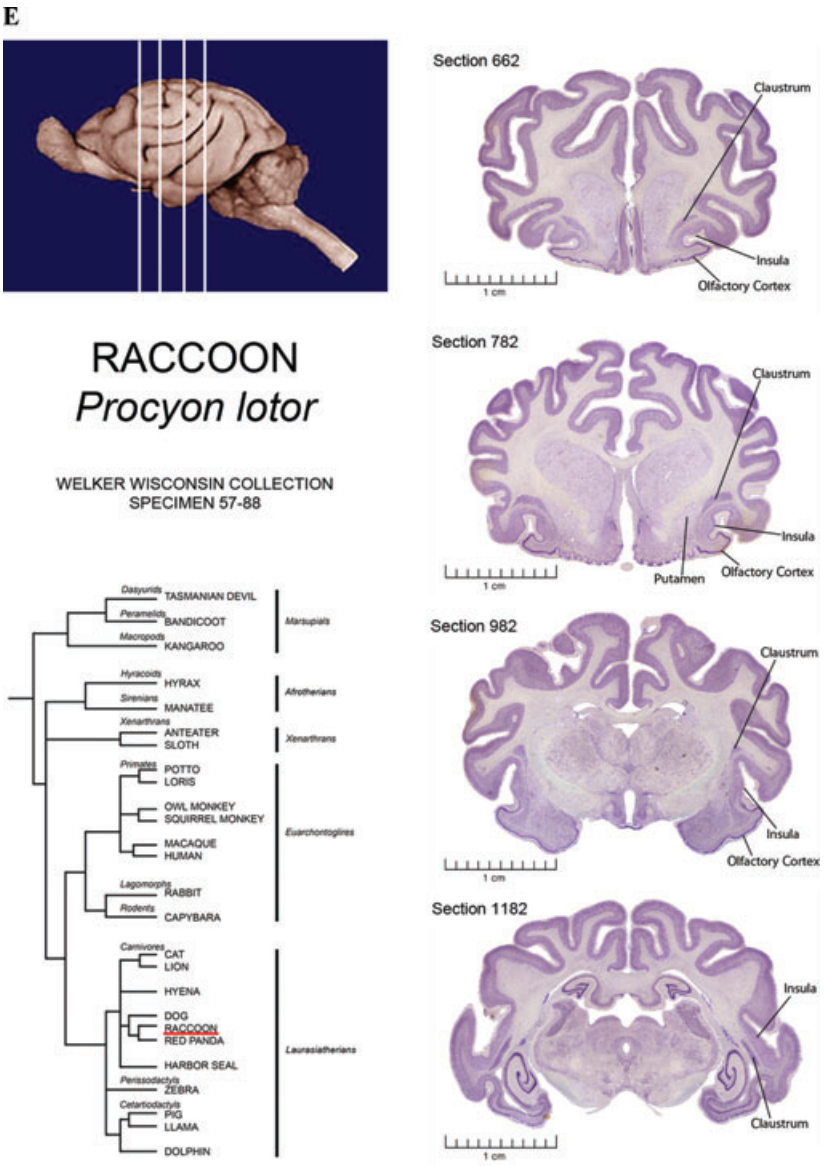


Figure 7. *Continued*. (E) Section 982 of the brain of this caniformian and procyonid raccoon shows a very slender claustral column reaching from the claustral root at the rhinal sulcus up to a very slender carnivoran triangular top. Anteriorly, in section 782, the claustral head widens into a broad club shape when it leaves the confines of the bounding putamen on its medial side. The rhinal sulcus shows an insulating expansion of its superior operculum, accompanied by the claustrocortex of the club-shaped claustral head. In the posteriormost section, 1182, the fundus of a sulcus expands into a sort of circular sulcus, forming a classically insulated region of claustrocortex at its base, reminiscent of the genuine insulas seen in primates, with operculations above and below the sulcal opening.

superior operculum. The raccoon has a very distinct and well-formed insula. For example, compare the insular region of the raccoon with that of the related red panda, which has similar brain and body sizes, yet shows a very small insula. The harbor seal has the largest insula, as well as the largest brain, of any of the carnivores that we examined. All of these carnivores have well-developed and distinct extreme capsules.

Claustrum. The carnivore claustrums we examined follow a relatively consistent pattern. In the

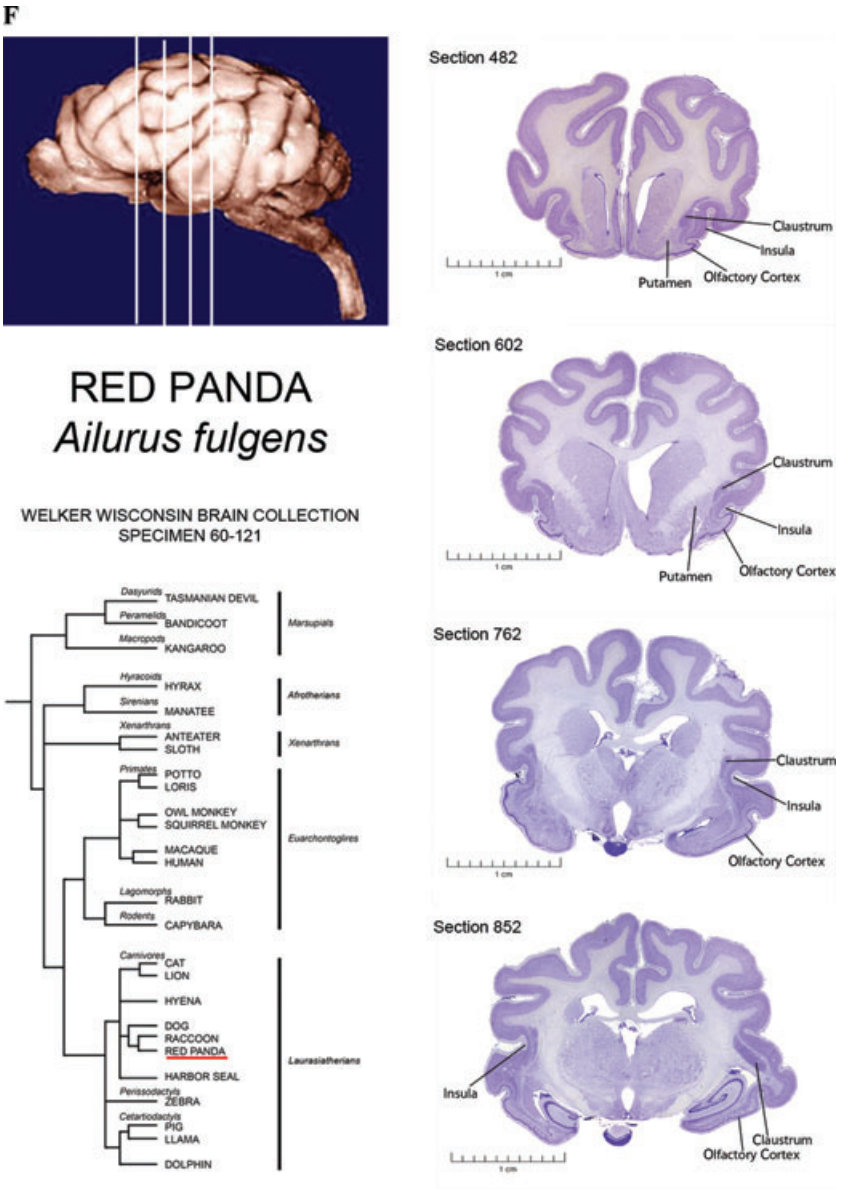


Figure 7. *Continued.* (F) This caniformian red panda has a somewhat smaller, less-elaborated brain than does its fellow procyonid raccoon. This relative simplicity also appears in the morphology of the claustro-insular region. The enlarged superior claustral head has a simple club shape, with an inferior stem leading to the root near the endopiriform entities. The enlargement takes place, as usual, when the claustrum is freed from medial impingement by the more inferior small putamen. A distinctive feature is the widening at the posterior end of the claustrum, under a claustrocortex that neatly occupies the space beneath a gyral formation that resembles the protoinsular gyri seen in the anteater and the hyrax.

most anterior sections, it has a fat hook-like shape. This form is found in each carnivore we examined, although the long axis of the hook takes a horizontal form in the cat and lion. In all of the carnivores, the claustrum accompanies a small anterior insula, wrapping around it and reaching back laterally to meet with endopiriform cell groups. As we move posterior, the expansion of the putamen and amygdalar complex leads to a flattening of the inferior claustrum, leading to a nearly vertical claustrum

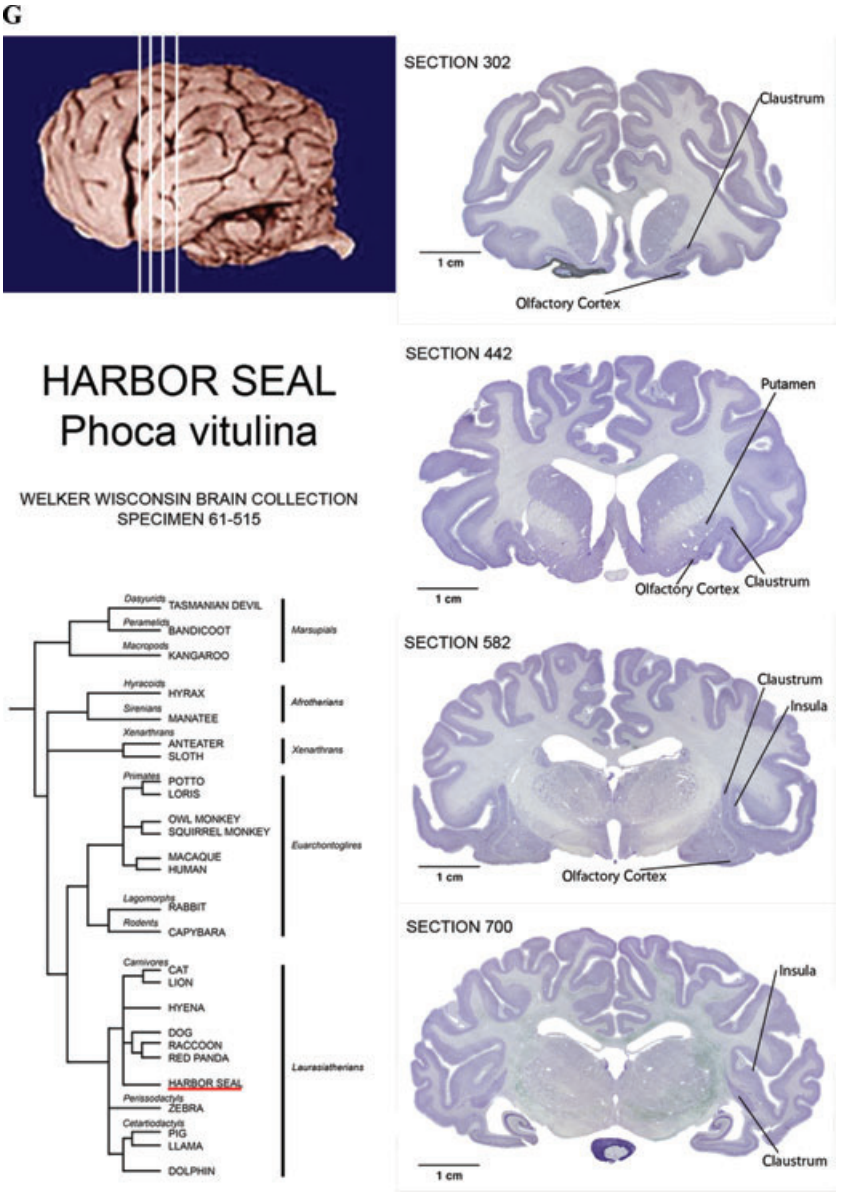


Figure 7. *Continued.* (G) The aquatic caniformian pinniped clade is represented here by this harbor seal. For a brain of its size and level of gyrencephaly, the claustral and insular morphology is rather simple. Anteriorly, in sections 301 and 442, a modest sized claustrum adheres closely to the margins of the rhinal sulcus with little evidence of an extreme capsule. In section 582, the long claustro-endopiriform root defies separation into claustral and endopiriform subdivisions using any criterion available in these sections. The cortex overlying this region is, however, insulated by a long overlying temporal gyrus. In the posterior section, 700, the widest part of the claustrum is rather far from the cortical tissue and much closer to the wall of the inferior horn of the lateral ventricle, a very unusual configuration among the mammals that we have examined. [Correction statement added after online publication on 27 May 2011: The final sentence of the caption for Figure 7G was deleted after publication.]

in each species. A very common feature was the presence of a large, triangular claustral expansion, at the level of the amygdala, spanning the sulcal fundus where the insular cortex meets its superior operculum in the superior claustrum. This can be seen in the cat section 718, lion section 1000, dog section 1220, and raccoon section 982, but is not found in the harbor seal. In each case, this feature

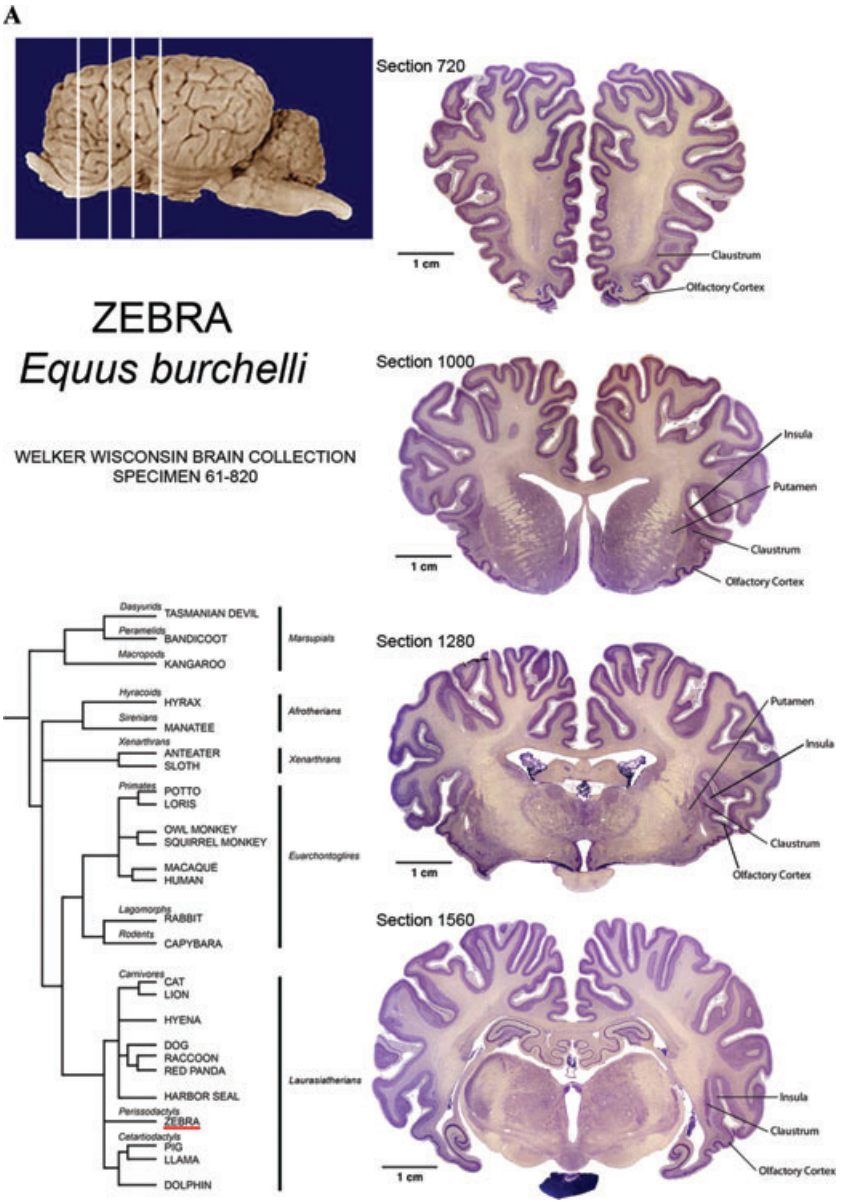


Figure 8. Perissodactyl and cetartiodactyl brains. (A) Our sole representative of the smallish extant order of perissodactyls, the brain of a zebra, shows features of interest in each of our survey sections. The anteriormost section, 720, features a gyrated claustrocortex. Four neighboring gyri overlie a claustral layer with points of cellular extensions extending up into the white cores of the neighboring gyri, as we have seen for individual gyri in hyrax, primate, and carnivore brains. In the next posterior section, 1000, the claustrum has congealed into a carnivore-like triangle shape, with its lateral apex pointing into the core of a single gyrus, bounded below by the rhinal sulcus. This claustrocortical gyrus is buried, that is to say, insulated, from above by a long operculating gyrus across a deep sulcus. A short claustral root connects firmly with endopiriform cell groups at the level of the fundus of the rhinal sulcus. In the remaining two posterior sections, 1280 and 1560, the now more-flattened claustrum adjoins a claustrocortical surface that is completely insulated by the superior operculum, forming a genuine insula. In section 1560, although not a part of the current investigation, note the fantastic folding of the hippocampal cornu ammonis on the upper portion of the cerebrum.

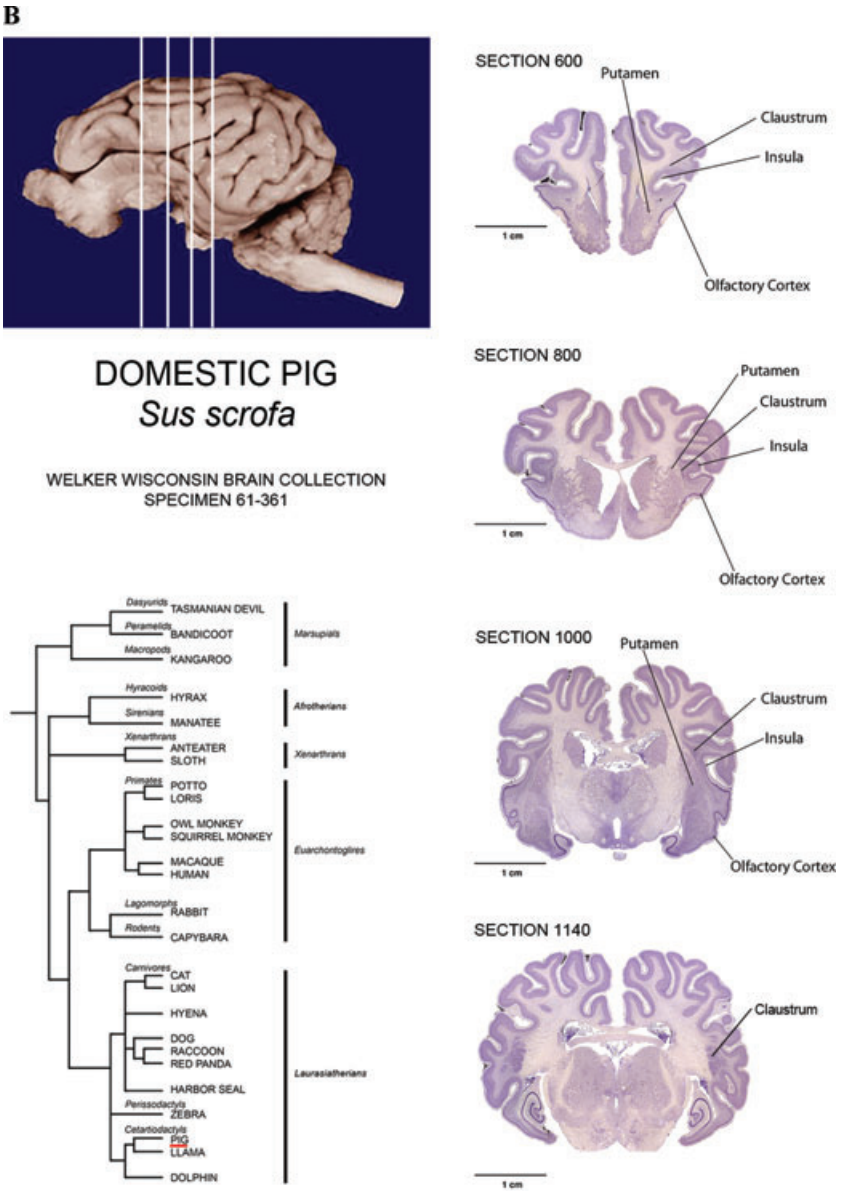


Figure 8. *Continued.* (B) The first of our cetartiodactyl specimens, a domestic pig, shows even more fascinating elaborations of its claustral and insular formations. At the anteriormost level in section 600, the anterior end of the claustrum underlies an operculating gyrus that forms the upper bank of a very deep rhinal sulcus. The lower bank is formed by the olfactory cortex, which has folded out to become a large gyral formation unlike any we have observed in our other mammalian specimens. Next, more posteriorly, in sections 800 and 1000, a completely insulated gyrus has arisen in the middle of the fundus of the rhinal sulcus, and the claustrum underlying this claustrocortical gyrus has expanded into a triangular shape; the lateral apex of the triangle, as in the carnivores, extends into the core of the overlying gyrus. The overlying gyrus is completely buried under the banks of the neighboring gyri, forming a genuine insula. The claustrum continues superior to its triangular center, as a thin lamina of cells curling around the sulcal fundus into the wall of the operculum formed by the superior-lying gyrus. Thus, the insula is completely floored by claustrocortex, but the claustrocortex continues beyond the upper limit of this insula. From the lower apex of the claustral triangle, the usual long tail of cells forms the claustral root, reaching past the rhinal sulcus to a distinctive large mass of endopiriform and para-amygdalar cells. In the most posterior section, 1140, is the most interesting feature of all. Deep to the cortex of an ectosylvian gyrus-containing subgyri, there is a very large mass of claustral cells, with striations suggesting lobulation or interruption by passing myelinated fiber groups. The remarkably large accumulation of claustral cells at the posterior end of the pig claustrum suggests that pigs, particularly in their laboratory-sized minipig form, may be advantageous subjects for experimental studies of claustral physiology.

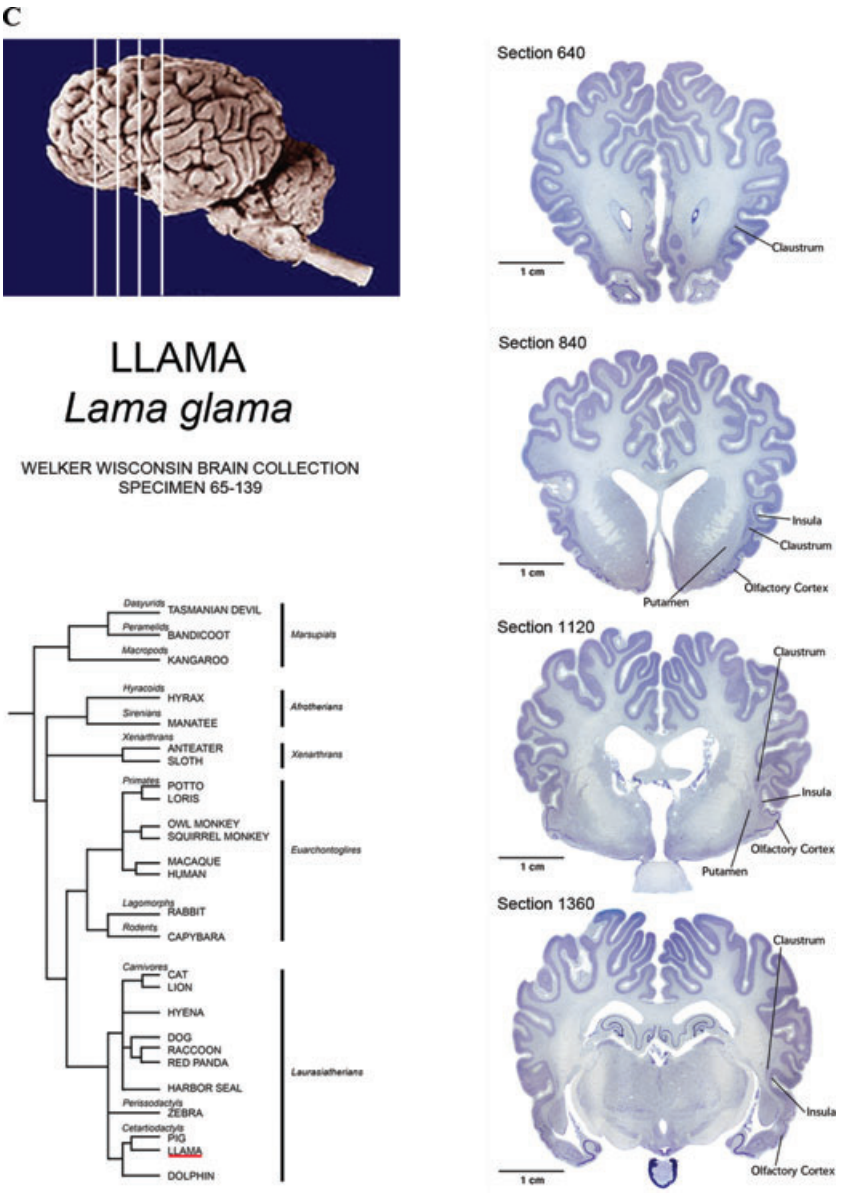


Figure 8. *Continued.* (C) In its most anterior regions, seen in section 640, the claustrum of this llama appears abruptly as a long-scalloped lamina, with thin extensions reaching far into the cores of four neighboring claustrocortical gyri. These gyri are on the lateral surfaces of the frontal portions of the cerebral hemispheres, anterior to the frontal poles of the putamen. In the next most posterior of the sample sections, 840 and 1120, the main portion of the claustrum is a thick, shorter cell group, triangular in appearance like that seen in some carnivores, but still maintaining thin extensions into long gyral cores. The sulci between these gyri have become deep and some have lateral extensions at their fundi, forming insulated patches of claustrocortex beneath the widened sulcal bases. This array has the appearance of a gyrencephalic insula without the overall covering of a lobular operculum. The claustral root into the endopiriform region, as in the carnivores, extends down from the inferior aspect of the claustral triangle. In the posteriormost section 1360, a posteriormost wisp of claustrum lies deep to the fundus of a sulcus, a hindmost bit of insulated claustrocortex.

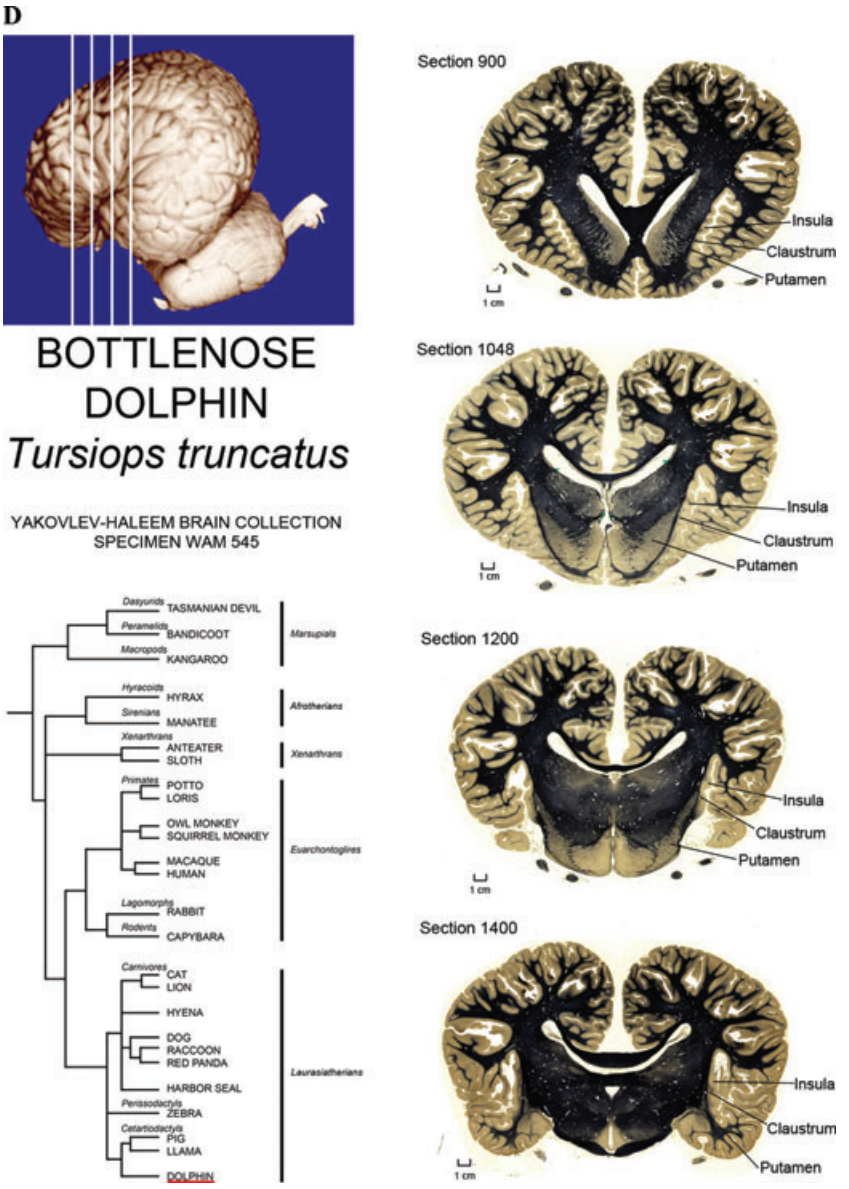


Figure 8. *Continued.* (D) As in the llama, here in section 900, the anteriormost appearance of the claustrum in this dolphin is a long, thin, scalloped lamina extending into overlying gyral cores, but between the gyral extensions the continuity of the claustrum is often lost, leaving islets of claustral tissue marking the locations of the several overlying gyri. As in the llama, the number of claustrocortical gyri diminishes proceeding caudally through sections 840, 1120, and 1360, and the width of the claustral lamina increases. However, the overlying gyri in the dolphin are themselves thoroughly insulated by overlying frontal and temporal lobes. Section 900 provides an interesting all-in-one-section survey of the possible, and here, actual spatial relations that can and do exist between insular cortex and claustrocortex. Starting at the uppermost limit of the insular cortex, there is a region with no underlying claustrum: insular cortex that IS NOT claustrocortex. Below this region, there is a second region where there is a claustrum beneath the cortex: insular cortex that IS claustrocortex. Furthermore, in this region the upper portion consists of isolated islets of claustrum that become a continuous lamina at more inferior regions. Finally, at the inferior extremity of the claustrum, the claustrum extends beyond the insular cortex and comes to lie under cortex that is on the external surface of the brain, thus constituting claustrocortex that IS NOT insular cortex. As there is no olfactory bulb or tract in adult dolphins, it is difficult to identify olfactory cortex and endopiriform regions from the morphological data that we have available. For this reason, we have not identified a claustral root leading to endopiriform cell groups. The stain used in these sections was the Loyez version of the Weigert hematoxylin stain for myelinated fibers.

is found superior and medial to the insula, extending into the superior insular operculum. Trailing this triangular claustral mass is a long, thin trail of cells reaching inferiorly and forming a cell bridge, the claustral root, which reaches periamygdalar endopiriform cell groups. The endopiriform group is particularly distinct in both the dog and raccoon. Finally, in its most posterior extent, the carnivore claustrum becomes slightly shorter and wider, and flanks the inferior insula and often stretches beyond the insula. The harbor seal is again different, as the posterior claustrum does not reach either the superior or inferior extent of this animal's large insula.

In most of the carnivores we examined, as we have seen in many mammals, the postero-superior insula is claustrum free. Besides this one small area, however, in carnivores, the insular cortex is all claustrocortex. However, the claustrum often escapes the insula at some point. At the level of the triangular claustral expansion, the claustrum commonly extends into the superior insular operculum. Despite this, in the carnivores there is a close correspondence between the extents of insular and claustrocortex.

Perissodactyls and cetartiodactyls

This large and varied grouping consists of the relatively small group of large and herbivorous perissodactyls, and the much more extensive (and recently taxonomically merged) groups, artiodactyls and cetaceans. In these large-brained animals, the claustrum fills spaces between U-fibers connecting neighboring gyri, seen most dramatically in cetaceans, llama, and zebra (Fig. 8), and also some large primates (Fig. 5D and F) and carnivores (Fig. 7B, C, and D).

Insula. In our perissodactyl specimen, the zebra, we observed that the insula begins well posterior to the anterior origin of the claustrum. For most of the extent of this insula, there is no inferior operculum, and the cortex is covered by a larger superior operculum. The insula continues for quite a distance beyond the most posterior extent of the claustrum.

Each of the three cetartiodactyls we examined possesses a fully formed insula, and each has a distinct morphology. The llama brain and insula closely resemble those of the zebra. However, the zebra brain is larger overall and has a much larger insula, including an insular gyrus, while the llama insula is little more than an expanded sulcus. The pig insula is also distinctly formed and expands from a large rhinal sulcus. In its anterior extent, it features a large insular gyrus that expands posteriorly and folds to become the superior operculum of the posterior insula. The dolphin insula is very large and features many radial gyruses. The tremendous expansion of the temporal lobe in the dolphin has buried a large area of cortex.

Claustrum. The zebra claustrum is another example of the accumulation of claustral cells in insular gyruses, and shows a poor correspondence between the insular cortex and claustrocortex. Unlike the carnivores, kangaroo, capybara, and pig, the zebra claustrum does not wrap around the rhinal sulcus anteriorly. The claustrum is thick, and the inferior claustrum expands into the spaces between the U-fibers connecting the local gyri. Posteriorly, the superior portion of the claustrum thickens and forms a triangular mass that also expands into the gyral cores.

The claustrums of the cetartiodactyls that we examined are quite varied and interesting, perhaps due to the large brain sizes of these mammals. The llama claustrum begins anterior to the insula, and seems to invest several gyruses on the lateral surface of the brain, filling in the space between their U-fibers and the underlying fasciculus of the extreme capsule. Posteriorly, it forms a claustral head in the superior insular operculum. The dolphin claustrum is very thin and long, following the extended course of the dolphin insula. It appears to expand to fill the depths of the many dolphin insular gyruses. The dolphin claustrum also breaks into claustral islets. Perhaps these breaks in the continuity of the claustrum are due to large fiber tracts running from the large temporal lobe. The dolphin claustrum also ends far before the posterior extent of the insula.

One of the most unique and surprising claustrums in our survey is that seen in our pig. We found that this claustrum expands much farther along the cortex than any other that we observed. In the anterior portion it has a tail, reaching into the superior operculum. This tail is longer than the body of the claustrum proper. More posteriorly, the claustrum has a large, triangular accumulation of cells in the superior operculum, and another claustral tail lies deep to the cortex of the ectosylvian gyrus. Finally, in its most posterior sections, the pig claustrum widens dramatically, and

is crossed by many visible fiber bundles, giving it a striated or lobulated appearance. These features can also be seen in a published atlas of the pig brain.[32]

Discussion

The shape of the claustrum

The claustrum takes on a variety of shapes across the mammalian radiation; however, there are several general trends in this diversification. The shape of the adult claustrum is likely to be influenced by the initial distribution of claustral cells and the volumetric expansion of local structures, including the amygdala, striatum, and white matter tracts, along with cortical folding. When the amygdala and putamen are very close to the cortex, the claustrum takes a thin and sheet-like form. It bulges in areas where these structures are not present or are not located so proximally to the cortical surface. This suggests that the claustrum does not need to have a particular shape and does not leave an impression on, or otherwise influence the morphology of, neighboring groups of cells or fibers. Rather, it is these neighboring structures that shape a morphologically passive mass of claustral cells, which tend to hang together in whatever shape they are molded. This would explain the common presence of the anterior bulge in the claustrums of the marsupials, xenarthrans and hyrax. These bulges all occur anterior to the level of the striatum, and these claustrums all become thin in posterior sections close to the putamen. We believe that the substantial postnatal expansion of the putamen may lead to this morphology. A similar phenomenon can be seen in the carnivore radiation, where, in adults, a mass of claustral cells can be found above the level of the putamen. The expansion of the putamen may flatten the claustrum in its inferior extent, leaving the superior region much wider. In several of the large-brained animals, claustral cells can be found extending into the cores of the gyruses of the insular region. As the claustrum forms substantially earlier than any insular gyruses,[33] these cells may be pushed into the gyral cores by the expansion of fiber tracts and local subcortical structures. Unfortunately, very little work has been done toward elucidating the developmental timeline of subcortical structures in these many different animals. Knowledge of the order of formation and volumetric expansion of these brain areas would offer great insight into the validity of any hypothesis about the morphology of the claustrum.

Another factor that may influence the structure of the claustrum is the number of claustral cells arising from a given area of cortex during development. This may explain the claustral tails that are often seen adjacent to cortex superior to the body of the claustrum. These tails may be thin not because they have been compressed, but rather because there was a relative paucity of such claustral cells settling deep to this area of cortex during development.

Terminology: the claustrocortex *and* insular cortex

The claustrocortex is not synonymous with the insular cortex. Of the animals that we examined with well-operculated insulas, several have claustrums that form well anterior to any operculation. The superior posterior portion of the insula is almost never claustrocortex. In the large-brained animals, the insula is much larger than the claustrum, usually leaving the posterior insula without claustrum. Many animals have portions of their claustral volume in one or more of the insular operculums. Finally, we were unable to recognize any clear effect of the structure of the claustrum on the structure of the insula. For this reason, we believe that the presence of the claustrum near the cortical surface may not play a significant role in the anchoring of the insular cortex during cortical expansion.

Olfactory connections and the morphologies of the insula and claustrum

Olfactory connections, in contrast, could provide a restricting influence that leads this region to be folded into an isolated ("insulated") patch of cerebral cortex. The presence of the "claustral root," a group of claustral cells leading into the olfactory-related endopiriform cell group, around the fundus of the rhinal sulcus, observed in 25 of our 26 specimens of diverse branches of the mammalian radiation (the brain of the anosmotic dolphin is the exception), lends support to the concept of an olfactory dimension of commonality of the structure, development, and function of claustrum in all mammals.

Insulation and cortical folding

We found well-operculated insulas in many of the large-brained mammals, including the human,

macaque, llama, zebra, dolphin, and harbor seal, but not the manatee. Of the smaller animals, we observed substantial operculation in some, such as the new world monkeys, raccoon, and dog, while others remained unoperculated, such as the grey kangaroo, and still others had proto-insulas, such as the lion, capybara, and the xenarthran anteater and sloth. Of note, we observed that the operculation of the insular cortex is not strictly dependent on the expansion of a temporal lobe. Some insulas are completely engulfed by their temporal operculum, while others were covered entirely by the superior frontoparietal operculum (these latter include those of the capybara, pig, raccoon, and zebra).

The most consistent trend we found in the insulas was their continual occurrence where multilaminated cortex meets the olfactory allocortex. It may be that the mammalian parvolaminated olfactory allocortex cannot support or limits the mechanism of cortical expansion that causes extensive lobes, and the close association of the insular cortex with the olfactory allocortex restricts the gyrencephalic expansion of this region.

The claustrum is always continuous with endopiriform cell groups connected with olfactory cortex. The insula is always neighboring olfactory cortex and receives connections from it. This joint attachment to olfactory cortex could account for the general localization of insulated cortices and claustral formations near one another, even though functionally there may be no special relationship between the claustrum and the neighboring insulated cortex.

Conflicts of interest

The authors declare no conflicts of interest.

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